

The Standalone Standard Model Structure from First Principles

An AASC Derivation of Gauge-Product, Representation, Higgs/Bridge, Yukawa, Charge, Anomaly, Vacuum, and Parameter-Status Structure

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May 20, 2026

Abstract

This paper gives a dependency-compressed first-principles derivation of the standalone Standard Model structural tuple from AASC constraints. The derivation does not import the accepted Standard Model as a premise. It begins from admissibility, standing, reference, construction, and scope discipline, and proceeds through interface constraints, realized physical interior, gauge-product recovery, representation-response rigidity, Higgs/bridge fixation, Yukawa-realizability structure, charge-lattice rigidity, anomaly neutrality, vacuum compatibility, and mass-response/mixing slots. The conventional Standard Model appears as the recovered structural endpoint, displayed explicitly as a tuple rather than as a merely verbal role schema.

The claim is structural. It does not assert closure of the complete empirical parameter table. Gauge-coupling values, quark masses, CKM/PMNS numerical coordinates, Higgs potential, vev, scalar mass, self-coupling, weak-scale values, absolute mass scales, and other residual constants remain parameter-level targets unless a later module supplies the relevant admissible witness, exactness certificate, reconstruction rule, or zero-anchor derivation. The family-cardinality record $N_g = 3$ is included as part of the recovered standalone empirical SM structural envelope; it is not treated as a raw coupling-like or mass-like numerical parameter.

The trajectory terminology used by AASC is constructional rather than dynamical: physical dynamics enter through the realized physical modules, while AASC supplies the standing-preservation discipline under which those modules compose.

The quantitative significance is that the structural derivation opens a sectoral parameter program: the charged-lepton sector already supplies a downstream conditional one-anchor numerical readout with protected pole-boundary residual localization. Exact charged-lepton pole closure and zero-anchor absolute-scale derivation remain open. Thus the manuscript separates first-principles structural derivation of the standalone Standard Model from the downstream sectoral program of numerical parameter derivation.

Contents

1	Introduction: Dependency-Compressed Derivation	5
1.1	The distributed-derivation problem	5
1.2	Reading guide: core AASC terms	5
1.3	AASC-to-physics instantiation	6
1.4	Trajectory language and physical dynamics	6

1.5	Structural derivation versus parameter derivation	7
1.6	Main structural spine claim	7
1.7	Higgs-like bridge/fixation and Yukawa-compatibility status	10
1.8	Nonclaims	11
1.9	Proof-spine method	12
2	Dependency Ledger and Status Labels	13
2.1	Status labels	13
2.2	Module acronym table	13
2.3	Dependency ledger	13
2.4	Support theorem classes	15
2.5	Prior boundary baselines and later supersession modules	15
2.6	Graphical theorem-chain and compressed rigidity spine	16
2.7	Proof obligation ledger	17
3	Kernel Layer: Admissibility, Standing, Reference, and Construction	19
3.1	Minimal primitives	19
3.2	Kernel Necessity Theorem	19
3.3	Immediate consequences	19
3.4	Kernel certificate	20
4	Interface Layer: AMetric Boundary and Redescription Discipline	21
4.1	Metricity is not primitive	21
4.2	Admissible redescription	21
4.3	Boundary trace and endpoint discipline	21
4.4	Interface certificate	22
5	Anchor–Tensor–Skin Decomposition	23
5.1	The ATS separation	23
5.2	ATS theorem in spine form	23
5.3	Consequences for the charged-lepton arc	23
5.4	ATS certificate	23
6	Physical Realization Bridge	24
6.1	The bridge problem	24
6.2	Realized physical interior	24
6.3	Physics-facing bridge table	24

6.4	ZPE and renormalization discipline	25
6.5	Physical bridge certificate	26
7	Gauge and Role Carrier Layer	27
7.1	From physical interior to carriers	27
7.2	Gauge/role necessity	27
7.3	Gauge-product recovery and role-to-group identification	27
7.4	Standard Model relevance	28
7.5	Boundary trace as endpoint selector	29
7.6	Gauge/role certificate	29
8	Mass and Inertial Response	30
8.1	Mass is not primitive scalar content	30
8.2	One-anchor reconstruction	30
8.3	Mass-response certificate	30
9	Standard Model Structural Realization	31
9.1	Declared Standard Model structural envelope	31
9.2	Imported SR7 payload theorem	31
9.3	Recovered Standard Model structural tuple	32
9.4	Meaning of standalone Standard Model structure	34
9.5	Structural feature claim ledger	35
9.6	Standard Model Structural Spine Theorem	36
9.7	Why this is not reverse engineering	36
9.8	Structural certificate	36
10	Charged-Lepton Parameter-Sector Module	37
10.1	Module role	37
10.2	Role-cell residue	37
10.3	Square-root amplitude geometry	37
10.4	Boundary-transition carrier	38
10.5	One-anchor metric display	38
10.6	Protected pole-boundary correction	38
10.7	Protected trace remainder	39
10.8	Charged-lepton module certificate	39
11	Open Problems and Non-Closure Ledger	40

11.1	Structural closure status	40
11.2	Parameter closure status	40
11.3	Exact charged-lepton pole closure	40
11.4	Zero-anchor absolute charged-lepton scale	40
11.5	Downstream parameter-sector roadmap	40
12	Consolidated AASC-to-Standard-Model Theorem	42
13	Anticipated Objections and Responses	43
14	Conclusion	46
15	Operational Glossary	46
16	Module Certificates	48
16.1	Kernel module	48
16.2	Interface module	48
16.3	Physical realization module	48
16.4	ZPE/trace module	48
16.5	Gauge/role and SM structural module	49
16.6	Mass/numerical module	49
16.7	Charged-lepton module	49
17	Charged-Lepton Numerical Details	50
18	Open Problem Ledger	51
19	Claim-Level Support Ledger	52
20	Compression Map from Distributed Corpus to Spine	54
21	Reader Navigation Guide	54

1 Introduction: Dependency-Compressed Derivation

1.1 The distributed-derivation problem

The AASC derivation is distributed across several theorem arcs: kernel admissibility and standing, AMetric/interface discipline, realized physical interior, zero-point and renormalization standing, gauge/role structure, mass and numerical standing, Standard Model classification, mass-ratio reconstruction, charged-lepton residue closure, and pole-boundary residual analysis. The Standard Model is not inserted as an initial assumption in that chain; it is the recovered structural endpoint after the admissibility, physical-realization, gauge, representation, charge, anomaly, Higgs/bridge, Yukawa, vacuum, and mass-response constraints are composed. The distribution preserves local proof obligations, but it obscures the single derivational route unless the dependencies are stated in one place.

In this manuscript, a proof spine is a dependency-compressed derivation. It does not replace the support theorems. It states their compositional order, claim type, endpoint, and nonclaim boundary so that the structural derivation can be inspected as a single chain.

1.2 Reading guide: core AASC terms

The manuscript uses a compact AASC vocabulary. The following terms are used in their fixed technical sense throughout the paper.

Term	Working meaning in this manuscript
Standing	admissible identity-bearing status for a construction, object, role, or witness inside a fixed scope
Admissibility	fail-closed permission to carry standing under the declared scope, boundary, transport, and no-repair discipline
AMetric Boundary	the pre-metric admissibility boundary before coordinates, metrics, operators, probabilities, or numerical comparisons have standing authority
Anchor	a domain-fixing condition; moving it changes the construction problem or envelope
Tensor	load-bearing invariant, witness, compatibility relation, or role content; moving it changes standing-bearing structure
Skin	presentation surplus: notation, chart, coordinate, basis, gauge representative, normalization convention, or descriptive packaging
Trajectory	a finite fixed-scope construction history; it is not physical time evolution or Hamiltonian dynamics
Bridge	an admissible mediation between two standing-bearing layers, such as order-to-deformation mediation or quotient identification
Envelope	the declared boundary, scope, load, or empirical-instantiation regime within which a theorem is typed
Parameter-status discipline	the classification of numerical objects as structural, skin, boundary, calibration, conditional, ratio-level, forced, obstruction-only, open, or zero-anchor closed

1.3 AASC-to-physics instantiation

AASC is not used here as an additional force, field, dynamics, ontology, or hidden physical sector. It supplies the admissibility conditions under which a proposed physical construction can carry standing. The formalism becomes physics-facing only when the AASC kernel is instantiated on a realized physical domain D : non-null persistence, localized standing, field-carrier mediation, particle persistence classes, gauge-transition structure, vacuum support, deformation/mass response, boundary-detectable witnesses, and admissible transition histories.

A physical role is admitted only when it preserves the relevant standing quotient, transport ledger, boundary witnesses, and no-repair discipline inside D . Physical dynamics, amplitudes, Hamiltonians, scattering structures, renormalization flows, and transition kernels enter through the physical modules. AASC determines whether their use is admissible as a standing-preserving construction inside the declared scope.

1.4 Trajectory language and physical dynamics

AASC uses “trajectory” in a fixed-scope constructional sense. It does not replace physical dynamics with an epistemic, semantic, or merely formal surrogate. A trajectory is a finite same-scope construction history whose coherence is determined by standing preservation at each state. The trajectory layer supplies the bivalent path geometry of admissible construction: prefix closure, legal composition, identity-preserving invariance, tensor determination, skin neutrality, endpoint certification, and irrecoverability of failure under extension [5].

Physical dynamics enter only after a physical domain D is instantiated. Hamiltonian propagation, gauge transport, scattering amplitudes, mass-response behavior, RG flow, boundary transition kernels, and vacuum-sector dynamics are physical-domain structures. The AASC trajectory layer governs whether the constructional use of those structures preserves standing inside the declared scope. It does not assert that physical time evolution, Hamiltonian dynamics, or quantum amplitudes are merely construction trajectories.

Principle 1.1 (No physical-dynamics replacement). AASC trajectory theory constrains the admissibility of construction histories. It does not replace Hamiltonian evolution, gauge transport, scattering dynamics, RG flow, vacuum-sector QFT, or mass-response dynamics. Whenever such structures appear in the proof spine, they enter through the physical modules and are then tested for standing preservation, quotient stability, boundary-witness preservation, and no-repair discipline.

Bridge	Domain	Function	Nonclaim
ZPE quotient bridge	homogeneous vacuum-energy sector	identifies the renormalized-QFT quotient with AASC standing preservation for scalar vacuum assignments	does not solve cosmological-constant matching
Gauge-transition bridge	local physical presentation domains	forces boundary-fixed local/restriction groupoid organization for admissible gauge transitions	does not derive gauge couplings
Higgs/MGN bridge	SR7 order/deformation sector	mediates admitted order witnesses into admitted deformation/mass-response standing through the canonical bridge carrier	does not derive Higgs vev, mass, or potential values
Yukawa/MGN bridge	canonical bridge carrier	admits only coupling-witness structures over $\mathcal{B}_D$	does not derive Yukawa matrices
Neutrino reconstruction bridge	source-propagation-detection domain	instantiates the AASC kernel on a physical boundary-transition witness and recovers the admissible transition kernel	does not derive PMNS values

1.5 Structural derivation versus parameter derivation

A recurring source of misreading is the conflation of Standard Model structure with the complete empirical Standard Model parameter table. This paper separates them.

By “standalone Standard Model structure” we mean the admissible structural occupant: the realized physical interior, gauge-product and product-sector organization, representation-response roles, charge lattice, anomaly-compatible survivor structure, vacuum compatibility, mass-response slots, Higgs-like bridge/fixation carrier, mixing/phase slots, and the closure of same-scope structural rivals. This is the object derived by the proof spine.

By “parameter derivation” we mean the later problem of deriving the numerical values assigned inside that structure: gauge-coupling values, Yukawa eigenvalues, charged-lepton and quark pole masses, CKM/PMNS numerical coordinates, CP phases, Higgs potential, vev, scalar mass, self-coupling, weak-scale values, scale anchors, and dimensionless constants. These are not all closed in the present manuscript.

Failure to close every parameter value is not failure to derive the structure. It means that the structural occupant has been derived and that numerical instantiation remains a downstream program. The charged-lepton arc is included as a parameter-sector module in that program.

1.6 Main structural spine claim

Theorem 1.2 (Main Structural Spine Claim). *Given the AASC kernel results and the support-theorem outputs listed in Section 2, the admissible route from construction constraints to physical realization is module-certified at each marked transition:*

$$\begin{aligned} \text{Adm/Stand} &\rightarrow \text{interface discipline} \rightarrow \mathcal{I}_{\text{phys}} \rightarrow \text{gauge/role carriers} \\ &\rightarrow \text{mass-response organization} \rightarrow \text{SM}_{\text{struct}}. \end{aligned}$$

Here $\text{SM}_{\text{struct}}$ denotes the standalone Standard Model structural occupant: the admissible realization profile containing the gauge/product-sector structure, representation-response roles, charge lattice,

anomaly-compatible survivor ledger, vacuum-supported residual form, Higgs-like bridge/fixation carrier, Yukawa-realizability and coupling-slot ledger, mass-response slots, mixing/phase slots, and same-scope rival exhaustion. The theorem is a structural derivation claim, not a claim of complete numerical parameter-table closure.

Theorem 1.3 (Downstream Parameter-Program Claim). *After the standalone Standard Model structure is fixed, remaining numerical values form a downstream parameter-derivation program. A parameter value is promoted from boundary, calibration, ratio-level, conditional, or open status only when a later module supplies an admissible witness, exactness certificate, quotient collapse, reconstruction rule, or zero-anchor derivation. The charged-lepton arc is one parameter-sector module in this program; it does not exhaust the parameter program.*

This claim is deliberately phrased as a spine claim. It asserts that the structural route is module-certified within the declared envelopes and that a nontrivial first numerical parameter-sector module has been constructed. It does not assert that every empirical constant in the Standard Model has already been derived.

Definition 1.4 (Standalone Standard Model structure). The standalone Standard Model structure is the admissible physical realization occupant consisting of:

1. a non-null realized physical interior;
2. standing-preserving gauge and product-sector carriers;
3. representation-response roles for matter sectors;
4. a rigid charge lattice up to admissible normalization/sign redescription;
5. anomaly-compatible survivor structure;
6. gauge-vacuum compatibility and vacuum-supported residual form;
7. Higgs-like bridge/fixation carrier and mass-response coupling slots;
8. mixing-angle and phase slots as admissible quotient/holonomy data where certified;
9. same-scope rival exhaustion under the declared admissibility envelope.

It is a structural object, not a completed table of empirical numerical values.

Definition 1.5 (Parameter-derivation program). The parameter-derivation program is the downstream numerical program that begins after the standalone Standard Model structure is fixed. Its targets include numerical gauge couplings, Yukawa eigenvalues, pole masses, mixing angles, CP phases, Higgs potential, vev, scalar mass, self-coupling, weak-scale values, scale anchors, and dimensionless constants. A parameter result is admitted only at the status certified by its module: forced, ratio-level, one-anchor, boundary-calibrated, conditional, obstruction-only, open, or zero-anchor closed.

Criterion 1.6 (Structure/parameter separation). A datum belongs to the structural Standard Model tuple when changing it changes the admissible object class, role ledger, witness family, operator-availability ledger, transport ledger, or anomaly/vacuum survivor ledger. A datum belongs to the parameter program when changing it changes a value inside the fixed structural occupant without changing those ledgers. Thus gauge product form, product-sector roles, representation-response slots, charge lattice, anomaly cancellation, vacuum compatibility, Higgs-like bridge/fixation carrier, mass-response coupling-slot/Yukawa-compatibility ledger, mass-response slots, mixing/phase slots, and the family-cardinality envelope record are structural. Gauge-coupling numerical values, Yukawa eigenvalues, pole masses, CKM/PMNS numerical coordinates, CP phase values, Higgs potential, vev, scalar mass,

self-coupling, weak-scale values, and absolute scale anchors are parameter-level unless a later exactness module promotes them.

Definition 1.7 (Recovered Standard Model structural tuple). The recovered standalone Standard Model structural tuple is

$$\Sigma_{\text{SM}} = (G_{\text{SM}}, \mathcal{R}_{\text{SM}}, H_{\text{br/fix}}, \mathcal{O}_Y, \mathcal{Q}_{\text{SM}}, \mathcal{A}_{\text{anom}}, \mathcal{V}_{\text{SM}}, \mathcal{M}_{\text{slot}}, \Theta_{\text{mix}}, N_g) / \simeq_{\text{Adm}} .$$

Here G_{SM} is the conventional presentation of the recovered product-sector role as $SU(3)_c \times SU(2)_L \times U(1)_Y$. This notation is used after the Lie-type, product-sector, representation-response, Abelian charge, and anomaly-ledger modules identify the admissible survivor with the conventional Standard Model gauge-product presentation up to admissible redescription. $\mathcal{R}_{\text{SM}}$ is the chiral representation-response skeleton; $H_{\text{br/fix}}$ is the Higgs-like bridge/fixation carrier required for order-to-deformation mediation and mass-response compatibility; $\mathcal{O}_Y$ is the Yukawa-realizability and coupling-slot ledger ($\mathcal{O}_{Y,u}, \mathcal{O}_{Y,d}, \mathcal{O}_{Y,e}; \mathcal{O}_{\nu,\text{status}}$), where the neutrino entry is envelope/sector-dependent and no Dirac, Majorana, Weinberg, or sterile-sector option is silently imported. $\mathcal{O}_Y$ records structural operator availability and coupling-witness topology over the canonical bridge carrier; it is not a numerical Yukawa matrix. $\mathcal{Q}_{\text{SM}}$ is the rigid charge-lattice ledger; $\mathcal{A}_{\text{anom}} = 0$ denotes the anomaly-neutral survivor ledger; $\mathcal{V}_{\text{SM}}$ is the vacuum-compatible residual normal form; $\mathcal{M}_{\text{slot}}$ and Θ_{mix} are the mass-response and mixing/phase slots; and $N_g = 3$ is the family-cardinality record of the standalone Standard Model envelope. The quotient by $\simeq_{\text{Adm}}$ removes skin-level gauge, basis, notation, normalization, and redescription choices.

Field role	Conventional SM presentation	Structural status
Q_L	$(3, 2)_{1/6}$	structural left-quark representation-response slot
u_R	$(3, 1)_{2/3}$	structural up-type singlet response slot
d_R	$(3, 1)_{-1/3}$	structural down-type singlet response slot
L_L	$(1, 2)_{-1/2}$	structural lepton-doublet response slot
e_R	$(1, 1)_{-1}$	structural charged-lepton singlet response slot
$H_{\text{br/fix}}$, conventionally H	$(1, 2)_{1/2}$, the conventional one-doublet presentation of the bridge/fixation carrier in the declared SM-load skin	structural bridge/fixation carrier; potential, vev, scalar mass, self-coupling, and weak-scale values downstream

Tuple entry	Primary support module	Certified content	Nonclaim boundary
G_{SM}	GRR/PSM/SR	gauge-product survivor profile	numerical gauge couplings
$\mathcal{R}_{\text{SM}}$	PSM/SR/charge-rigidity	chiral representation-response skeleton	Yukawa values or flavor fits
$H_{\text{br/fix}}$	MGN3 + mass-fixation support	bridge/fixation carrier and order-to-deformation mediation	Higgs potential, vev, mass, self-coupling
$\mathcal{O}_Y$	L1 Yukawa equivalence + MGN4	operator availability, Yukawa realizability, coupling topology, role-rank discipline	numerical Yukawa matrices, masses, CKM/PMNS
$\mathcal{Q}_{\text{SM}}$ $\mathcal{A}_{\text{anom}} = 0$	charge-lattice rigidity anomaly-ledger/SR	discrete charge lattice anomaly-neutral survivor ledger	g_1, g_2, e values UV completion
$\mathcal{V}_{\text{SM}}$	PSM/SR/ZPE discipline	vacuum-compatible residual form	cosmological constant prediction
$\mathcal{M}_{\text{slot}}$ Θ_{mix}	R9/MGN/CNF MGN/CNF	mass-response slots mixing/phase slots	all pole masses CKM/PMNS numerical values
$N_g = 3$	residual/family ledger	structural family-cardinality envelope record	same-mechanism derivation as charge/anomaly

Remark 1.8 (Family cardinality is structural, not a mass/coupling parameter). The family-cardinality record is part of the recovered standalone Standard Model structural tuple because changing it changes the representation-family ledger, mixing-coordinate dimension, Yukawa-compatibility structure, and admissible boundary bookkeeping. The support theorem set classifies the mechanical derivation of that cardinality as envelope-level rather than as a raw internal parameter calculation. That classification does not demote the Standard Model structure; it fixes where the cardinality lives in the structural tuple. The derivation fixes the admissible location and structural role of $N_g = 3$ inside the standalone tuple; it does not reclassify N_g as a mass-like or coupling-like numerical parameter, nor does it claim that the integer three is obtained by the same mechanism as charge-lattice rigidity.

1.7 Higgs-like bridge/fixation and Yukawa-compatibility status

Definition 1.9 (Higgs-like bridge/fixation carrier). The tuple entry $H_{\text{br/fix}}$ denotes the structurally required bridge/fixation carrier over the SR7 skeleton. It is the canonical carrier by which admitted order standing mediates into admitted deformation and mass-response standing. In MGN3 notation this carrier is represented by

$$\mathcal{B}_D = (W_D^{\text{br}} / \sim_D^{\text{br}}, \text{Sig}^{\text{br}}, \beta_D^{\text{red}}, \rho^{\text{br}}),$$

not by a raw scalar coordinate, vev-like value, potential coefficient, field normalization, empirical scale, Yukawa term, mass matrix, or mixing datum.

Remark 1.10 (Higgs support and non-import boundary). The Higgs-like bridge/fixation support proves structural mediation and fixation status, not premature Higgs phenomenology. MGN3 establishes bridge survivability, canonical carrier reduction, bridge role-rank discipline, no vev selector, no potential

selector, no normalization rank inflation, no Yukawa import, no mass-matrix import, and no mixing-angle import. The separate mass-fixation support states the same point in physical language: once gauge-invariant SM bookkeeping closes while stable mass parameters are required, an observable Higgs-like fixation excitation is structurally unavoidable. This is a fixation/standing claim, not an ontic “mass-giving” claim.

Status 1.11 (What is and is not claimed about the Higgs sector). This manuscript claims the Higgs-like bridge/fixation carrier required for the recovered structural tuple. It does not claim, in this step, derivation of the Higgs potential, vev value, scalar mass, scalar self-coupling, weak mixing angle, gauge-boson masses, fermion masses, or numerical Yukawa matrices. Where conventional Standard Model notation writes the electroweak scalar doublet $H \sim (1, 2)_{1/2}$, the structural derivation records the admissible bridge/fixation slot and the operator-compatibility ledger needed for mass-response. The conventional one-doublet presentation is the declared SM-load skin for that bridge/fixation carrier; the Higgs potential, vev, scalar mass, self-coupling, and weak-scale numerical values remain downstream unless a separate module certifies them.

Definition 1.12 (Yukawa-realizability and coupling-slot ledger). The tuple entry $\mathcal{O}_Y$ is not the numerical Yukawa matrix table. It is the Yukawa-realizability and coupling-slot ledger over the canonical bridge carrier. In the charged sectors it records the structural availability and compatibility status of the up-type, down-type, and charged-lepton Yukawa slots. It also records the MGN4 coupling-witness carrier

$$\mathcal{Y}_D = (W_D^Y / \sim_D^Y, \text{Sig}^Y, \gamma_D^{Y, \text{red}}, \rho^Y),$$

which is the standing-bearing survivor of Yukawa-like notation after bridge, representation, charge, anomaly, residual, order, deformation, quotient, and scale-transport filters are imposed. In the neutrino sector it records an explicit status variable $\mathcal{O}_{\nu, \text{status}}$: renormalizable Dirac, Majorana, effective Weinberg, sterile-sector, or absent options are not silently assumed. Any such option requires its own envelope declaration or downstream module. Numerical Yukawa matrices, fitted Yukawa values, mass eigenvalues, CKM/PMNS coordinates, physical phases, and hierarchy exponents are downstream parameter-sector objects, not entries of $\mathcal{O}_Y$.

Theorem 1.13 (Yukawa realizability as structural operator availability). *Within the minimal chiral load with the conventional one-doublet bridge skin, complete Yukawa-operator realizability is a structural transport condition rather than optional phenomenological data. Under admissible chiral-gauge/broken-phase transport, anomaly-freedom and complete Yukawa realizability are equivalent tensor requirements for the minimal chiral Standard Model load. A charge/representation assignment that fails to admit the required Yukawa operators leaves non-transportable massless chiral remnants or requires forbidden compensating load. Consequently $\mathcal{O}_Y$ belongs to the structural tuple as operator-availability and coupling-slot topology, while the numerical Yukawa values remain downstream.*

1.8 Nonclaims

The following nonclaims are part of the theorem typing. They distinguish structural derivation from parameter-table closure.

Nonclaim 1.14 (Scope of the master spine). This paper does not claim:

1. that every empirical Standard Model numerical parameter is derived here;
2. that the charged-lepton arc exhausts the parameter-derivation program;
3. that exact charged-lepton pole closure has been completed;

4. that the absolute charged-lepton mass scale has been derived without an anchor;
5. that quark masses, CKM values, PMNS values, Higgs potential, vev, scalar mass, scalar self-coupling, weak-scale values, or gauge-coupling numerical values are closed in this manuscript;
6. that the Standard Model Lagrangian is postulated and then reverse-engineered into AASC;
7. that metricity, fields, vacuum energy, or coordinates are primitive;
8. that observed m_μ or m_τ are used as construction inputs;
9. that external QFT formulas enter untyped; when used, they enter only through explicit physical-interface or calculation modules.

Remark 1.15 (What is claimed). The structural claim is not weakened by these nonclaims. The paper claims derivation of the standalone Standard Model structure. The nonclaims only deny that structural closure is identical to full numerical parameter-table closure.

1.9 Proof-spine method

The method of the paper is modular. Each section imports a support result, states its spine role, gives the theorem or proposition needed downstream, and closes with a module certificate. Each certificate records:

input output support used by status.

This is stricter than an overview. It is designed so a reader can locate any disputed step in the support theorem set and determine whether the step is proved, conditional, achieved, or open.

2 Dependency Ledger and Status Labels

2.1 Status labels

The paper uses four status labels.

Definition 2.1 (Status labels). • **Proved**: established by a support theorem and imported without rederivation.

- **Module-supported**: established by a named support arc; the present paper uses the result as a module output.
- **Conditional**: depends on a stated admission theorem, bridge certificate, or role-order certificate.
- **Open**: exact obstruction identified but not closed.

2.2 Module acronym table

For external readability, the major module labels are fixed here. Later sections use these labels as shorthand for support arcs rather than as unexplained jargon.

Acronym/module	Expansion	Role in this manuscript
AASC	admissible construction under constraints	kernel legality framework: admissibility, standing, reference, construction, scope
AMetric	AMetric boundary/interface layer	blocks primitive metricity before admissible physical fixation
ATS	anchor–tensor–skin	separates standing sources, load-bearing carriers, and display variables
CNF	constructive numerical forcing	rules for admissible numerical solver/certificate structure
MGN	mass-generation / numerical-readiness arc	mass, scale, coupling, and bridge constraints used by numerical modules
MR	mass-ratio arc	ratio/shape, finite-residue, and one-anchor reconstruction support
R1–R12	Realized Physics arc	physical interior, fields, particles, gauge, mass, vacuum, time, and causality bridge
ZPE	zero-point energy arc	renormalization/standing rules for vacuum offsets and relational boundary response
CLM–CLR	charged-lepton carrier arc	charged-lepton residue, square-root carrier, and one-anchor readout
CLS	charged-lepton residual arc	protected pole-boundary improvement and exact trace-remainder obstruction

2.3 Dependency ledger

Layer	Imported module	Output used in spine	Status
Kernel	AASC kernel: non-degenerate construction and kernel of admissibility, structure of admissibility, bivalence, standing identity closure	admissibility, standing, reference, construction discipline	proved / module
Interface	AMetric boundary, closure by exhaustion, boundary trace, invariant endpoint selection	no primitive metricity; scope discipline; redescription legality; finite endpoint discipline	proved / module
ATS	Anchor–Tensor–Skin decomposition	fixed-domain distinction between standing anchor, load-bearing tensor, and skin display	proved / module
Physical interior	Realized Physics R1–R12	non-vacuous physical persistence, locality, fields, particles, vacuum, causality, physical interior	module supported
ZPE/RG	ZPE arc	absolute vacuum offsets blocked; relational boundary-conditioned response allowed	proved / module
Gauge/role	boundary trace, R8, Gauge Realization, gauge-overlap closure, gauge-coupling moduli	admissible role carriers, gauge-product recovery, and gauge-coupling nonclaim boundary	module
Mass response	R9, MGN1–MGN8, MR arc	mass as deformation/inertial response; numerical standing; one-anchor metric reconstruction	module supported
SM structure	Standard Model classification, Structural, Gauge Realization, PSM, and SR1–SR7	standalone Standard Model structural occupant under AASC constraints	module- supported structural derivation
Charged-lepton carrier	CLM–CLR v4.6	2/9 role-cell residue, Z_3 -circulant carrier, first parameter-sector one-anchor readout	conditional / achieved
Pole residual	CLS v2.11	conditional protected pole-boundary residual-localization module and protected trace-remainder obstruction	partial / open

2.4 Support theorem classes

The support theorems are grouped as follows. The PDFs bundled with this manuscript are the minimal load-bearing support set, not the full AASC library. Their inclusion indicates that the present manuscript depends on their module outputs; it does not imply that every detail of those papers is rederived here. The list is curated by dependency role: kernel legality, interface discipline, physical realization, gauge realization, pre-Standard-Model bridge, Standard Model rigidity, numerical forcing, exactness, and charged-lepton readout.

- **Kernel and interface:** non-degenerate construction and the kernel of admissibility, structure of admissibility, bivalence, AMetric boundary, standing identity closure, closure by exhaustion, invariant endpoint structure, boundary trace fixation [1, 2, 3, 6, 4, 7, 8, 9].
- **ATS and metric discipline:** anchor–tensor–skin decomposition, metric fixation as representational choice, no global metric fixation [10, 11, 12].
- **Physical realization:** Realized Physics R1–R12 [32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43].
- **Gauge realization:** transition groupoid, connection as standing-preserving transport, curvature/holonomy obstruction, representation admissibility, charge-witness rigidity, anomaly obstruction, and gauge-vacuum compatibility [85, 86, 87, 88, 89, 90].
- **Pre-Standard-Model bridge:** Lie-type gauge realization, product gauge-sector compatibility, representation specialization, Abelian charge-witness normalization, anomaly-ledger algebraization, and vacuum-compatible residual normal form [19, 20, 21, 22, 23, 24].
- **Standard Model rigidity:** SR1–SR7, culminating in Standard Model realization rigidity in the admissible gauge arena [25, 26, 27, 28, 29, 30, 31].
- **Prior classification baseline:** Standard Model occupant classification, fixed-load interior closure, and gauge-coupling parameter non-collapse [13, 14, 15].
- **ZPE and renormalization standing:** non-predictivity of absolute zero-point vacuum energy and preservation of standing under renormalization [44, 45, 46].
- **Numerical readiness and mass ratios:** CNF4–CNF9, MGN3–MGN8, MR2/MR4/MR6/MR7 [50, 52, 53, 54, 55, 58, 62, 63, 64, 65, 66, 67, 68, 69, 70].
- **Exactness:** exactness criteria, quotient-singleton promotion, transport neutralization, scale/normalization exactness, calibration removal, endpoint/boundary exactness, discrete holonomy exactness, cross-sector exactness, and computable extraction [71, 72, 73, 74, 75, 76, 77, 78, 79].
- **Charged-lepton numerical reduction:** CLM, CLN, CLQ, CLR v4.6, CLS v2.11 [80, 81, 82, 83, 84].

2.5 Prior boundary baselines and later supersession modules

Earlier AASC Standard-Model papers classified several quantities as boundary, envelope, or externally fixed data within their then-declared residual scopes. Those results are retained as no-selector baselines. They establish that a parameter, cardinality, or residual value cannot be forced by a same-scope operation unless a later module supplies new admissible structure.

The later realized-physics, gauge-realization, PSM, Standard Model rigidity, numerical-readiness, constructive numerical-forcing, mass-ratio, charged-lepton, and exactness arcs do not erase those earlier baselines by assertion. They supersede them only where they provide one of the following:

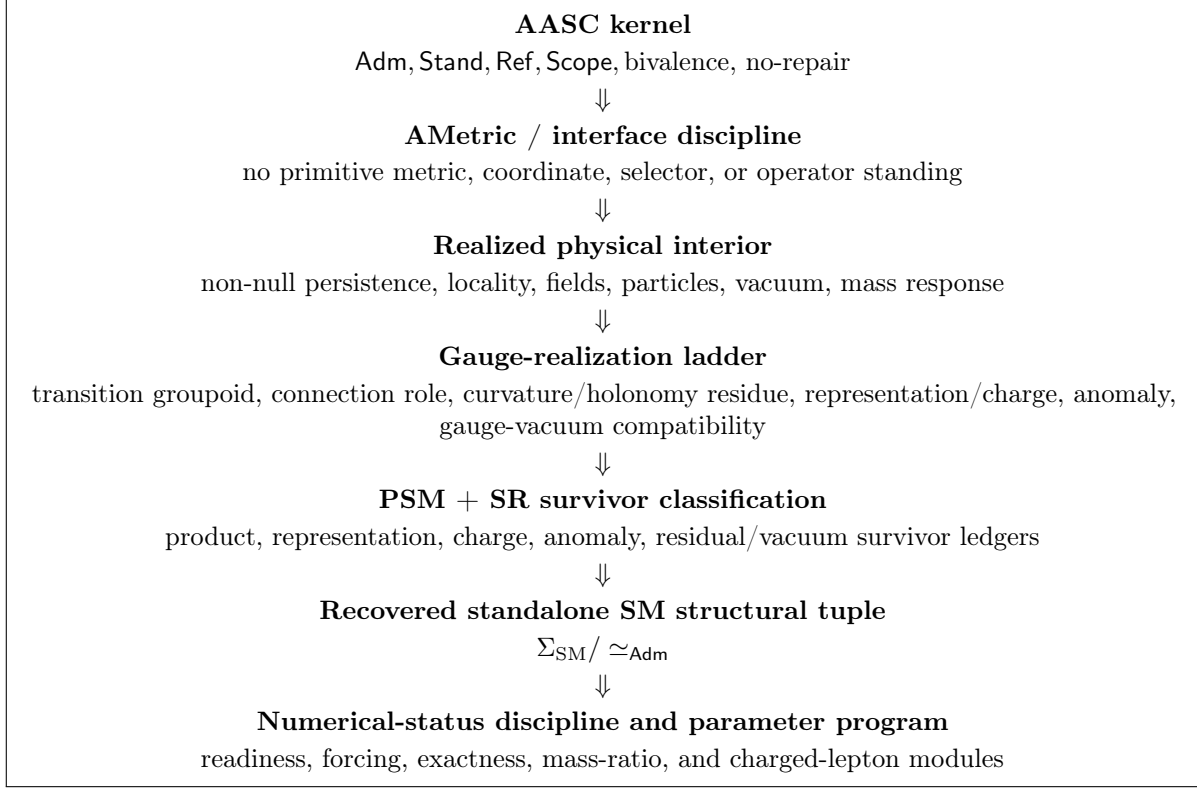
1. a new admissible witness;
2. a new carrier or transport structure;
3. a quotient collapse;
4. a role-rigidity theorem;
5. an exactness certificate;
6. a reconstruction rule;
7. a zero-anchor derivation.

Where no later module supplies such structure, the earlier boundary or EXT-FIX classification remains in force.

Layer	Earlier baseline	Later status in this spine
SM occupant	Classification / fixed-load closure	Strengthened by realized-physics, gauge-realization, PSM, and SR arcs into standalone SM structural realization.
Gauge couplings	Raw chart-numeric values not forced	Still not claimed as fully numerically derived. Only invariant/overlap/role status is used.
Charge lattice	Fixed-scope charge rigidity	Retained and strengthened as structural closure support inside the gauge-realization/SR sequence.
Generation count	Previously boundary/EXT-FIX in residual scope	Included in Σ_{SM} as the standalone SM family-cardinality envelope record $N_g = 3$; not treated as a raw numerical parameter.
Mass/Yukawa values	Boundary or residual status in older scope	Charged-lepton sector opened as first parameter module; other sectors remain downstream.
PMNS/CKM values	Reconstruction/slot status, not full numerical closure	Not claimed as fully derived here.

2.6 Graphical theorem-chain and compressed rigidity spine

The proof route can be read at two resolutions. The first is the global theorem-chain from the AASC kernel to the parameter program:



The decisive rigidity step is the second, physics-facing resolution. The Standard Model gauge product is not assumed at the first gauge step. It appears after the gauge role is forced into a transition architecture, then into transport comparison, curvature/holonomy residue, representation and charge admissibility, anomaly neutrality, and gauge-vacuum compatibility:

$$\begin{aligned}
&\text{boundary-fixed gauge role} \rightarrow \text{Gpd}_{\partial}(D) \rightarrow \nabla_{\partial D} \rightarrow \Omega_{\partial D} \\
&\rightarrow \text{Rep/Charge} \rightarrow \mathcal{A}_{\text{anom}} = 0 \rightarrow \text{GaugeVac}_D \rightarrow G_{\text{SM}}.
\end{aligned}$$

Composed with the PSM/SR survivor results, this chain recovers the conventional presentation $SU(3)_c \times SU(2)_L \times U(1)_Y$ as the gauge-product component of Σ_{SM} . The chiral representation table, Higgs/bridge carrier, Yukawa-realizability ledger, charge lattice, anomaly ledger, vacuum-compatible residual form, mass-response slots, mixing/phase slots, and family-cardinality envelope record then complete the recovered structural tuple. Thus the paper asks the reader to audit a finite chain of typed rigidity claims rather than to accept the Standard Model as an imported premise.

The corresponding transition certificates are: kernel necessity; interface theorem; physical-realization theorem; gauge-realization and PSM bridge; Standard Model rigidity; numerical-readiness/exactness discipline; and charged-lepton carrier/pole-boundary modules.

2.7 Proof obligation ledger

The following table records the minimum proof obligations that a critical reader should audit. When a support theorem uses a named theorem, the named result is listed; where theorem numbering varies by source version, the named theorem or section anchor is used instead of a page number.

Spine step	Theorem/certificate	Source anchor	State
Kernel legality	Kernel necessity for admissibility-by-standing	Non-Degenerate Construction and Kernel of Admissibility; Structure of Admissibility; Bivalence; Standing Identity Closure	proved / module
Metric non-primitivity	AMetric boundary and metric-fixation discipline	AMetric Boundary; Metric Fixation; No Global Metric Fixation	proved / module
Scope discipline	no illicit transport; standing redescription	Closure by Exhaustion; Boundary Trace; Structura	proved / module
Physical interior	realized persistence, locality, fields, particles, gauge, mass, vacuum, causality, physical interior	Realized Physics R1–R12	module
Gauge role	gauge-role induction from boundary-trace-fixed presentation equivalence	R8 Gauge Structure from Boundary Trace Fixation	module
Mass role	mass/inertia role induction from deformation-resistance under field-carrier mediation	R9 Mass, Inertia, and Persistence Compression	module
SM structural occupant	role-rigid standalone SM realization profile	Gauge Realization; PSM; SR1–SR7; Classification; Structura	structural closure
Charged-lepton parameter module	typed charged-lepton role-cell residue and square-root amplitude endpoint	CLM; CLQ; CLR v4.6	conditional / module
One-anchor display	one-anchor metric reconstruction as skin import, not zero-anchor scale derivation	CLN; MR6; ATS; Metric Fixation	achieved / module
Pole residual	conditional protected pole-boundary residual-localization module and protected trace-remainder obstruction	CLS v2.11	partial / open
Zero-anchor scale	leaf-relative scale carrier requirement	ATS; Metric Fixation; MGN/MR scale bridge	open

3 Kernel Layer: Admissibility, Standing, Reference, and Construction

3.1 Minimal primitives

The proof spine begins before geometry, fields, particles, masses, Hilbert spaces, Lagrangians, or coordinates. The primitives used by the support theorems are:

Adm, Stand, Ref, Scope, construction.

Admissibility is the legality constraint on construction and redescription. Standing is the global binary property by which a construction-bearing object survives admissible redescription. Reference is standing-preserving admissible redescription. Scope determines the domain over which those constraints are evaluated. These notions are not metric, dynamical, or field-theoretic.

3.2 Kernel Necessity Theorem

Theorem 3.1 (Kernel Necessity). *A non-degenerate construction system with stable reference requires admissibility-by-standing. Without admissibility, construction has no legality criterion; without standing, reference has no preserved object; without reference, construction cannot produce a stable target.*

Proof spine. The support kernel establishes that coherent construction is not merely a sequence of inscriptions. A construction must preserve the possibility of a stable object across admissible redescriptions. That preservation is standing. If a move destroys standing, it cannot be repaired by later success, because success would then be measured against a target whose reference has already failed. Hence admissibility is not an optional evaluative layer; it is the constraint under which construction remains construction rather than unconstrained symbol production [1, 2, 3, 4]. $\square$

As established in *Non-Degenerate Construction and the Kernel of Admissibility*, every non-degenerate regime of admissibility-assessable, identity-preserving construction on a fixed domain instantiates the AASC kernel by necessity; the kernel is not generated by such systems, but is what makes their non-degeneracy possible. In the realized physical envelope declared in this manuscript, that kernel instantiation licenses the downstream support chain: AMetric/interface discipline, the realized physical interior I_{phys} , the gauge/role carrier ladder, PSM/SR survivor classification, and the recovered standalone Standard Model structural tuple Σ_{SM} . Thus the Standard Model spine is a forced consequence not of an unconstrained kernel alone, but of the kernel composed with the stated physical-realization and standing-preservation modules.

3.3 Immediate consequences

The kernel supplies the following downstream prohibitions:

1. no silent redescription;
2. no post-selection repair of standing failure;
3. no parameter insertion by numerical convenience;
4. no elevation of a display coordinate to standing without a witness;
5. no use of downstream success as proof of upstream admissibility.

These prohibitions later block fitted charged-lepton corrections, absolute vacuum-energy predictions, global metric fixation, and untyped pole-boundary counterterms.

3.4 Kernel certificate

Module output	admissibility-by-standing constraint engine
Dependency	non-degenerate construction and kernel of admissibility; structure of admissibility; bivalence; standing identity closure
Used by	interface layer; realized physical interior; numerical forcing; pole-boundary residual discipline
Status	proved / module

4 Interface Layer: AMetric Boundary and Redescription Discipline

4.1 Metricity is not primitive

The AMetric boundary is the first major physical-looking consequence of the kernel, although it is not yet physics. It says that metricity does not enter as a primitive. Coordinates, distances, scales, local fields, and dynamical variables are not the starting point. They are possible representational carriers after admissible fixation.

Theorem 4.1 (AMetric Interface Theorem). *Before an admissible physical interior is licensed, no metric, coordinate, dynamical, field, or scale descriptor may function as primitive standing. Such descriptors may enter only as representational carriers within a licensed scope.*

Proof spine. If metric descriptors were primitive, they would be exempt from the admissibility and standing tests that govern construction. The AMetric boundary rules out that exemption. A metric carrier can be introduced for coordination after the relevant scope is fixed, but it has no global primitive standing. This follows from the AMetric boundary, metric fixation, and no-global-fixation results [6, 11, 12]. $\square$

4.2 Admissible redescription

A redescription

$$X \rightsquigarrow Y$$

is admissible only if standing and reference are preserved within the declared scope. A redescription is not legalized by producing a useful coordinate, a concise formula, or an empirically attractive value. The support theorems use this rule to fail-close candidate structures.

Proposition 4.2 (No illicit scope transport). *No leaf-relative, local, or skin-level descriptor may be exported into a global or boundary-level role unless an independent standing-preserving transport theorem licenses that export.*

This proposition protects the later proof from two common errors: treating a metric skin as a physical anchor, and treating an interior scheme coordinate as a global trace/pole response.

4.3 Boundary trace and endpoint discipline

The interface layer also supplies boundary trace fixation and invariant endpoint selection. These are crucial when an interior construction leaves a finite family of candidates. A boundary trace can fix an endpoint only if the trace itself is stable, admissible, scope-compatible, and not post-selected from the target value [9, 8].

4.4 Interface certificate

Module output	AMetric boundary, redescription legality, no illicit scope transport, endpoint discipline
Dependency	AMetric boundary; closure by exhaustion; invariant endpoint selection; boundary trace fixation
Used by	physical realization; gauge/role carriers; ZPE; metric scale discipline; charged-lepton pole residuals
Status	proved / module

5 Anchor–Tensor–Skin Decomposition

5.1 The ATS separation

The proof spine uses the Anchor–Tensor–Skin decomposition as a fixed-domain discipline. The terms are operational.

Definition 5.1 (Anchor, tensor, skin). An *anchor* is a standing source that licenses a domain or scale of comparison. A *tensor* is the load-bearing carrier that transports or organizes standing content. A *skin* is the representational display in which a tensor output is expressed.

The distinction is essential for the Standard Model spine. It prevents treating MeV masses, pole schemes, coordinate metrics, or vacuum offsets as primitive anchors. It also prevents reducing the physical derivation to language about appearances.

5.2 ATS theorem in spine form

Theorem 5.2 (Fixed-Domain ATS Theorem). *In an admissible description, skin-level display cannot create standing. Standing must enter through an anchor and be transported through a load-bearing tensor. A skin may display a result; it cannot generate it.*

Proof spine. A skin is a coordinatized display. If it could create standing, any convenient coordinate could be elevated to physical status. That violates AMetric and standing-preservation discipline. Therefore the skin can only inherit standing from an anchor/tensor chain [10, 6, 11]. $\square$

5.3 Consequences for the charged-lepton arc

The electron pole mass used in the charged-lepton module is a skin-level metric import, not a zero-anchor derivation of the charged-lepton scale. The Z_3 -circulant carrier is a tensor-level mass-shape object. The remaining exact pole residual is a tensor-to-skin problem, not a license to fit the skin.

5.4 ATS certificate

Module output	anchor/tensor/skin separation
Dependency	ATS fixed-domain decomposition; AMetric boundary; metric-fixation discipline
Used by	physical realization, mass response, charged-lepton scale import, CLS pole-boundary tensor analysis
Status	proved / module

6 Physical Realization Bridge

6.1 The bridge problem

The kernel and interface layers are not yet physics. They establish what cannot be smuggled into a derivation. The physical-realization bridge asks how a physical interior can be admitted without making metricity or dynamics primitive.

The answer supplied by the Realized Physics arc is that a non-vacuous physical interior is admitted as boundary-licensed persistence and tensor organization. Physical structure enters as a constrained interior, not as a primitive ontology of spacetime points or fields.

6.2 Realized physical interior

Let $\mathcal{I}_{\text{phys}}$ denote the realized physical interior admitted by the support theorem set. It is not a bare set of events. It is a persistence-bearing interior in which locality, continuation, fields, particles, vacuum, time, and causality are progressively licensed under AASC constraints [32, 33, 34, 35, 36, 37, 38, 41, 42, 43].

Theorem 6.1 (Physical Realization Theorem). *AASC admits physical structure only as a boundary-licensed realized interior whose carriers preserve standing, reference, scope discipline, and admissible redescription. Physical structure therefore enters as constrained tensor organization rather than primitive metric ontology.*

Proof spine. R1 blocks physically null realization. R2–R5 build locality, continuation neighborhoods, metric-role comparison, and finite propagation as downstream persistence structures. R6 and R7 license fields and particles as persistence carriers/classes. R10–R12 complete the physical-interior statement: vacuum, time/causality, and physical interior are admitted only after continuation and persistence constraints. None of these results introduce metricity as primitive; they admit physical structure through constrained realization [32, 33, 34, 35, 36, 37, 38, 41, 42, 43]. $\square$

6.3 Physics-facing bridge table

The realized-physics module is used in this spine through specific physical roles, not through a generic declaration that “physics exists.” The following table records the physics-facing content imported by the spine.

Physical role	Support source	Spine use
Non-null realization	R1	prevents physical realization from collapsing into empty reconstruction
Locality/domain structure	R2–R4	supplies localization, local domains, and metric-role comparison without primitive metricity
Finite transport	R5	supplies bounded perturbation transport before gauge specialization
Fields	R6	treats fields as distributed persistence carriers, not primitive ontology
Particles	R7	treats particles as localized persistence classes
Gauge structure	R8	supplies boundary-trace-fixed presentation equivalence as minimal gauge role
Mass/inertia	R9	supplies deformation-resistance role for mass and inertia
Vacuum	R10 and ZPE arc	treats vacuum as minimal concrete persistence regime and blocks absolute vacuum-offset predictions
Time/causality	R11	treats irreversible continuation ordering as the physical role later represented by time/causality
Physical interior closure	R12	closes the realized physical interior used by later gauge/mass/SM modules

6.4 ZPE and renormalization discipline

The ZPE arc supplies a complementary rule. Absolute zero-point vacuum energy is not a cosmological prediction merely because a formal sum can be written. Invariance and renormalization discipline block absolute vacuum offsets as standing physical predictions. Relational, boundary-conditioned, response quantities may carry standing if properly witnessed [44, 45, 46].

Remark 6.2 (ZPE quotient bridge). The zero-point-energy arc supplies a physics-bound bridge between AASC and standard QFT. In the homogeneous vacuum-energy sector, the QFT requirement that a scalar assignment descend to the finite-renormalization quotient coincides with the AASC requirement that a scalar assignment preserve standing under admissible redescription. The absolute zero-point coefficient fails both tests for the same reason: it is a global additive offset without an invariant relational witness. Thus the ZPE result is not an additional vacuum dynamics. It is a quotient-identification bridge between renormalized QFT practice and AASC standing discipline.

Proposition 6.3 (Vacuum/trace admissibility). *Vacuum and trace-like terms are admissible only as relational, boundary-conditioned, witness-bearing response objects. They are inadmissible as absolute offsets or scheme coordinates promoted to physical predictions.*

This proposition later governs the charged-lepton protected trace remainder. The finite pole-boundary remainder cannot be an arbitrary finite counterterm. It must be a relational, leaf-relative trace response.

6.5 Physical bridge certificate

Module output	boundary-licensed realized physical interior; vacuum/trace standing discipline
Dependency	Realized Physics R1–R12; ZPE arc; AMetric interface
Used by	gauge/role structure, mass response, Standard Model structural class, CLS trace/pole analysis
Status	module supported

7 Gauge and Role Carrier Layer

7.1 From physical interior to carriers

Within $\mathcal{I}_{\text{phys}}$, physical objects must be carried by structures that preserve standing under allowed transformations. That is the origin of the gauge/role layer in the spine. Gauge-like structure is not inserted as a Lagrangian premise. It is the admissible organization of role-stable transport and boundary-compatible carriers.

7.2 Gauge/role necessity

Theorem 7.1 (Gauge/Role Carrier Theorem). *In the realized physical interior, admissible physical organization must be represented by role-stable, boundary-compatible transport carriers. Gauge-like structure is the admissible organization of those carriers under standing-preserving redescription.*

Proof spine. The physical interior licenses fields and particles as persistence carriers/classes rather than primitive local data. Transport across admissible descriptions must preserve role and boundary trace. R8 identifies gauge structure from boundary trace fixation. The Gauge Realization arc then supplies the transition groupoid, connection, curvature/holonomy obstruction, representation/charge admissibility, anomaly obstruction, and gauge-vacuum compatibility. The PSM arc specializes this bridge toward Lie-type gauge realization, product-sector compatibility, representation algebra, Abelian charge normalization, anomaly algebraization, and vacuum-compatible residual normal form [37, 38, 39, 9, 15, 85, 86, 87, 88, 89, 90, 19, 20, 21, 22, 23, 24]. Therefore the gauge/role layer is module-certified by stable transport and boundary-compatible realization, not postulated as a convenient symmetry story. $\square$

7.3 Gauge-product recovery and role-to-group identification

Gauge structure is not introduced as an ontic symmetry postulate. It enters the spine as the admissible redescription and transition carrier required for standing-preserving local physical presentation. The Gauge Realization sequence sharpens this carrier through the ladder

$$G_D \Rightarrow \text{Gpd}_{\partial}(D) \Rightarrow \nabla_{\partial D} \Rightarrow \Omega_{\partial D} \Rightarrow \text{Rep/Charge} \Rightarrow \mathcal{A}_{\text{anom}} = 0 \Rightarrow \text{GaugeVac}_D.$$

Only after this ladder is composed with the PSM, SR, and charge-lattice modules is the conventional Standard Model gauge-product presentation recovered:

$$G_{\text{SM}} = SU(3)_c \times SU(2)_L \times U(1)_Y.$$

Thus the gauge product is not a primitive group assumption in the proof spine. It is the conventional physics-facing notation for the admissible gauge-product survivor after transition organization, transport comparison, curvature/holonomy residue, representation response, charge-witness rigidity, anomaly neutrality, and gauge-vacuum compatibility have been imposed.

Theorem 7.2 (Imported gauge-product recovery payload). *The gauge-realization support supplies a staged recovery of the gauge component of Σ_{SM} . The transition layer forces a boundary-fixed local/restriction groupoid organization $\text{Gpd}_{\partial}(D)$; the transport layer supplies connection-like standing-preserving comparison; the holonomy layer supplies curvature/transport-residue obstruction; the representation and charge layer supplies admissible response and charge-witness rigidity; the anomaly*

layer supplies standing-failure obstruction; and the vacuum layer supplies gauge-vacuum compatibility. Composed with the PSM and SR modules, these results recover $SU(3)_c \times SU(2)_L \times U(1)_Y$ as the Standard Model gauge-product presentation up to admissible redescription and declared normalization conventions.

Remark 7.3 (Gauge structure is representational, not ontic). The proof spine rejects an ontic reading of gauge symmetry. Gauge symmetry organizes admissible local descriptions and transition equivalences; it is not an additional primitive substance or an independently existing multiplicity of worlds. This is why the structural theorem can treat gauge as load-bearing redescription and transport structure while refusing to treat gauge-related presentations as distinct physical occupants [18, 85].

Remark 7.4 (Gauge-product structure versus gauge-coupling values). The structural theorem concerns the gauge-product survivor, representation-response ledger, charge/hypercharge lattice, anomaly-neutrality condition, and gauge-vacuum compatibility. It does not derive the numerical values of g_1, g_2, g_3 , e , α , running coupling values, or any unification-scale relation. Those quantities remain parameter-sector or enlarged-domain targets unless a later module supplies a strongly isolating dynamical, boundary, or exactness certificate [16, 17, 15].

Layer	Support module	Certified content	Nonclaim
Gauge role	R8 and realized-physics bridge	boundary-fixed presentation equivalence preserving field and particle roles	no specific Lie group yet
Transition organization	GRR1	local/restriction groupoid layer $\text{Gpd}_\partial(D)$	no bundle or SM group at this layer
Transport comparison	GRR2	connection-like standing-preserving comparison role	no numerical gauge potential
Loop residue	GRR3	curvature/holonomy obstruction role	no Yang–Mills dynamics here
Rep./charge	GRR4; charge rigidity; SR3–SR4	representation-response and charge-witness rigidity	no gauge-coupling values
Anomaly	GRR5; SR5	anomaly cancellation as standing-failure obstruction	no UV-completion prohibition
Vacuum compatibility	GRR6; PSM6; SR6	gauge-vacuum compatibility and breaking admissibility	no Higgs numerical sector
Product survivor	PSM; SR; classification	$SU(3)_c \times SU(2)_L \times U(1)_Y$ as recovered SM gauge-product presentation	no g_1, g_2, g_3 derivation

7.4 Standard Model relevance

The support theorems supply two SM-facing layers. The older classification and Structura papers provide the baseline occupant/interior closure and no-hidden-load discipline. The later Gauge Realization, PSM, and SR arcs supply the realization-rigidity bridge: they recover the Standard Model gauge-product presentation and then close product, representation, charge, anomaly, and residual/vacuum survivor ledgers into the standalone Standard Model structural occupant [13, 14, 85, 86, 87, 88, 89, 90, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31].

7.5 Boundary trace as endpoint selector

The gauge/role layer also supplies a later tool: when a role carrier has internal ambiguity, stable boundary trace can fix an endpoint. This is used in gauge-coupling standing, charged-lepton carrier admission, and pole-boundary trace analysis [9, 39, 15].

7.6 Gauge/role certificate

Module output	gauge/role carrier architecture; Gauge Realization/PSM bridge; boundary-trace endpoint discipline
Dependency	R6–R8; Gauge Realization; PSM; boundary trace fixation; gauge-coupling uniqueness
Used by	standalone SM structural theorem, mass-response layer, charged-lepton carrier and residual tensor
Status	module supported

8 Mass and Inertial Response

8.1 Mass is not primitive scalar content

Mass does not enter the spine as a primitive scalar attached to a particle. The R9/MGN/MR modules treat mass as deformation response, inertia, compression, and metric-relative readout inside the realized physical carrier system [40, 58, 62, 63, 64, 65, 66, 67, 69].

Theorem 8.1 (Mass Response Theorem). *In the realized physical interior, mass is an admissible readout of role-dependent deformation/inertial response. A numerical mass displayed in a metric unit is a skin-level representation of an upstream response carrier.*

Proof spine. R9 identifies mass/inertia/persistence compression as physical response rather than primitive numerical label. MGN1–MGN8 and the MR arc then specify numerical standing, scalar/order witnesses, bridge rigidity, Yukawa compatibility, eigenstructure, phase standing, hierarchy/ratio rigidity, numerical readiness, shape classes, finite residues, and one-anchor metric reconstruction. The spine imports the result: mass ratios and shapes can become prediction-ready only when their carriers have standing; metric display requires an anchor or scale bridge [40, 58, 62, 63, 64, 65, 66, 67, 68, 69, 70]. $\square$

8.2 One-anchor reconstruction

One-anchor reconstruction is not a failure of the spine. It is the proper expression of the ATS and metric-fixation rules when the shape carrier is derived but absolute scale has not been zero-anchor derived. If the dimensionless shape is

$$[r_e : r_\mu : r_\tau],$$

then one-anchor metric display gives

$$m_i^{(e)} = m_e^{\text{obs}} \frac{r_i}{r_e}.$$

This imports a skin-level scale. It does not import the muon or tau targets.

8.3 Mass-response certificate

Module output	mass as deformation/inertial response; shape/ratio readiness; one-anchor metric reconstruction
Dependency	R9; MGN numerical readiness arc; MR mass-ratio arc; ATS metric discipline
Used by	charged-lepton numerical module; pole-boundary residual analysis; zero-anchor scale open problem
Status	module supported

9 Standard Model Structural Realization

9.1 Declared Standard Model structural envelope

The Standard Model structural step is relative to a declared admissibility envelope, but the terminal target is not merely a loose “SM-like” family. The terminal target is the standalone Standard Model structural occupant: the structural realization profile left after the AASC kernel, physical-realization bridge, gauge-realization modules, PSM specialization modules, representation-response modules, charge-rigidity modules, anomaly-ledger modules, vacuum-compatibility modules, and Standard Model rigidity modules have been composed.

The envelope is

$$E_{\text{SM}} = \{\text{realized physical interior; standing-preserving carrier transport; boundary-trace fixation; gauge/product-sector realization; representation-response roles; charge/anomaly/vacuum compatibility; mass-response readout; fixed-domain comparison; no new hidden load}\}.$$

Within this envelope, the later gauge-realization, PSM, and SR modules strengthen the older classification/fixed-load baseline into a standalone Standard Model structural realization claim [85, 86, 87, 88, 89, 90, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 13, 14].

9.2 Imported SR7 payload theorem

Theorem 9.1 (Imported SR7 Payload). *SR7 supplies the capstone gauge-realization payload required by the structural derivation. Given the realized physical interior, Gauge Realization sequence, PSM1–PSM6 bridge, and SR1–SR6 survivor projections, the SR7 joint survivor object has a load-minimal realization class that is unique in the fixed SR7 comparison regime up to admissible role-equivalence. The survivor profile contains the product, representation-response, Abelian charge, charge-ratio, anomaly-neutral, and residual/vacuum roles required for the Standard Model structural tuple. Any candidate differing from that class is skin-equivalent, quotient-redundant, admissibly redescriptive, hidden-load expanding, larger-domain substitutive, repair-dependent, parameter-selected, downstream mass-generation data, or inadmissible.*

Remark 9.2 (Use of SR7 in this spine). SR7 is used as a structural survivor theorem, not as a numerical parameter theorem. It closes the admissible gauge-realization skeleton. The Higgs-like bridge/fixation, mass-response, Yukawa-compatibility, mixing/phase, and numerical-value layers are then typed by MGN, CNF, Exactness, MR, and sector-specific parameter arcs. This sequencing keeps the structural claim strong without moving unresolved numerical values into the structural proof.

Theorem 9.3 (Imported MGN3 Higgs/Bridge Payload). *MGN3 supplies the bridge payload required by the structural derivation. Over the SR7 skeleton, a Higgs-like bridge has standing only when it mediates from admitted order-witness classes to admitted deformation-witness classes while preserving the SR7 product, representation, charge, anomaly, residual/vacuum, deformation, and scale-transport ledgers. Its standing-bearing survivor is the canonical bridge carrier $\mathcal{B}_D$, not a scalar notation, vev coordinate, potential coefficient, normalization, empirical scale, Yukawa term, mass matrix, or mixing datum. Consequently $H_{\text{br/fix}}$ is a structural bridge/fixation entry of Σ_{SM} ; Higgs-sector numerical values remain parameter-level unless separately certified.*

Theorem 9.4 (Imported MGN4 Yukawa Payload). *MGN4 supplies the coupling-slot payload required by $\mathcal{O}_Y$. Given the canonical bridge carrier $\mathcal{B}_D$, a Yukawa-like numerical occurrence has standing only if it factors through an admitted coupling witness, preserves bridge-carrier compatibility, representation standing, charge standing, anomaly standing, residual normal form, order/deformation standing, quotient stability, and scale-transport discipline, and is not generated by field basis, flavor basis, normalization, scheme, empirical fit, texture ansatz, vev value, mass matrix, eigenvalue, mixing angle, physical phase, or hierarchy import. Its standing-bearing survivor is the canonical coupling carrier*

$$\mathcal{Y}_D = (W_D^Y / \sim_D^Y, \text{Sig}^Y, \gamma_D^{Y, \text{red}}, \rho^Y).$$

Thus Yukawa compatibility is structural operator availability and coupling topology over the bridge carrier, not a table of numerical Yukawa values.

Theorem 9.5 (Imported L1 Yukawa-realizability Payload). *Within the fixed minimal chiral Standard Model load and the conventional one-doublet bridge skin, anomaly-freedom and complete Yukawa-operator realizability are equivalent tensor requirements across the unbroken chiral-gauge and broken-phase bookkeeping skins. The apparent independence of anomaly/charge structure and Yukawa/mass structure is a skin illusion: if Yukawa operators are not realizable, transport to the broken-phase skin leaves massless chiral remnants or requires forbidden compensating load. Therefore Yukawa operator availability belongs to the standalone structural tuple, while the numerical values of Yukawa entries remain downstream parameter data.*

Remark 9.6 (Mass-fixation support). The mass-fixation support adds the physical interpretation of the MGN3 payload: within gauge-invariant SM bookkeeping, stable mass parameters require an admissible fixation structure with observable excitation. The result is not that the Higgs ontically causes mass. The result is that hidden or unobservable fixation would fail standing; an observable Higgs-like fixation excitation is structurally unavoidable once the closed SM bookkeeping requires stable mass parameters.

9.3 Recovered Standard Model structural tuple

The recovered object is displayed in conventional physics-facing form as

$$\begin{aligned} \Sigma_{\text{SM}} = (&SU(3)_c \times SU(2)_L \times U(1)_Y, \mathcal{R}_{\text{SM}}, H_{\text{br/fix}}, \mathcal{O}_Y, \mathcal{Q}_{\text{SM}}, \\ &\mathcal{A}_{\text{anom}} = 0, \mathcal{V}_{\text{SM}}, \mathcal{M}_{\text{slot}}, \Theta_{\text{mix}}, N_g = 3) / \simeq_{\text{Adm}} . \end{aligned}$$

This tuple is the standalone Standard Model structure recovered by the proof spine. The entries have the following statuses.

Tuple entry	Meaning in this spine	Status
$SU(3)_c \times SU(2)_L \times U(1)_Y$	conventional presentation of the recovered gauge-product survivor after transition, connection, curvature, representation, charge, anomaly, and vacuum-compatibility filtering	structure
$\mathcal{R}_{\text{SM}}$	chiral representation-response skeleton and matter-sector role ledger	structure
$H_{\text{br/fix}}$	Higgs-like bridge/fixation carrier required for order-to-deformation mediation and mass-response compatibility; not a vev/potential/value claim	structure/bridge slot
$\mathcal{O}_Y$	mass-response coupling-slot/Yukawa-compatibility ledger ($\mathcal{O}_{Y,u}, \mathcal{O}_{Y,d}, \mathcal{O}_{Y,e}; \mathcal{O}_{\nu,\text{status}}$); not the numerical Yukawa eigenvalue table	structure
$\mathcal{Q}_{\text{SM}}$	rigid charge-lattice ledger with admissible normalization/sign redescription	structure
$\mathcal{A}_{\text{anom}} = 0$	anomaly-neutral survivor ledger	structure
$\mathcal{V}_{\text{SM}}$	vacuum-compatible residual normal form	structure
$\mathcal{M}_{\text{slot}}$	mass-response slots; values remain sectoral parameter targets	structural slots
Θ_{mix}	mixing/phase slots; numerical coordinates remain downstream targets	structural slots
$N_g = 3$	family-cardinality record of the standalone Standard Model envelope	structural envelope record

The proof spine fixes $N_g = 3$ as part of the recovered standalone empirical SM structural envelope and proves that family cardinality is not a tunable same-scope parameter. It does not claim that the integer three is obtained by the same mechanism as charge-lattice rigidity or anomaly neutrality.

This table blocks the charge that the manuscript recovers only a role schema. The role language is the derivation medium. The recovered object is the displayed Standard Model structural tuple.

The conventional field-role presentation of $\mathcal{R}_{\text{SM}}$ is:

Field role	Conventional representation	Structural status
Q_L	$(3, 2)_{1/6}$	left-quark doublet representation-response slot
u_R	$(3, 1)_{2/3}$	up-type singlet response slot
d_R	$(3, 1)_{-1/3}$	down-type singlet response slot
L_L	$(1, 2)_{-1/2}$	lepton doublet representation-response slot
e_R	$(1, 1)_{-1}$	charged-lepton singlet response slot
$H_{\text{br/fix}}, \text{ conventionally } H$	$(1, 2)_{1/2}$, the conventional one-doublet presentation of the bridge/fixation carrier in the declared SM-load skin	structural bridge/fixation carrier; potential, vev, scalar mass, self-coupling, and weak-scale values downstream

Tuple entry	Primary support	Certified content	Nonclaim boundary
G_{SM}	GRR/PSM/SR	recovered gauge-product survivor $SU(3)_c \times SU(2)_L \times U(1)_Y$	no numerical gauge couplings
$\mathcal{R}_{\text{SM}}$	PSM/SR/charge rigidity	chiral representation-response skeleton	no Yukawa values
$H_{\text{br/fix}}$	MGN3; mass-fixation support	bridge/fixation carrier	no Higgs potential, vev, or mass value
$\mathcal{O}_Y$	L1; MGN4; Structura	operator availability and coupling-slot topology	no numerical Yukawa matrices
$\mathcal{Q}_{\text{SM}}$	charge-lattice rigidity	charge and hypercharge lattice	no g_1, g_2, e values
$\mathcal{A}_{\text{anom}} = 0$	GRR5; SR5; Structura	anomaly-neutral survivor ledger	no UV-completion ban
$\mathcal{V}_{\text{SM}}$	PSM6; SR6; ZPE discipline	vacuum-compatible residual form; absolute ZPE offsets excluded from standalone predictive status	no cosmological-constant prediction or boundary/matching solution
$\mathcal{M}_{\text{slot}}$	R9; MGN/CNF	mass-response slots	no full pole-mass table
Θ_{mix}	MGN/CNF	mixing and phase slots	no CKM/PMNS numerical values
$N_g = 3$	residual/family ledger	structural envelope record	not same-mechanism derivation as charge/anomaly

9.4 Meaning of standalone Standard Model structure

The terminal structural target is the standalone Standard Model structure. The target includes:

1. a realized physical interior with admissible carriers;
2. gauge-product recovery and product-sector organization;
3. representation-response roles for matter sectors;
4. rigid charge-lattice structure;
5. anomaly-compatible survivor ledger;
6. vacuum-supported residual structure;
7. Higgs-like bridge/fixation carrier and mass-response coupling slots;
8. mixing and phase slots where certified;
9. the family-cardinality envelope record $N_g = 3$;
10. admissible numerical-readiness and forcing status for parameter sectors;
11. same-scope rival exhaustion.

It does not include a completed derivation of every measured numerical parameter. The latter is the downstream parameter-derivation program.

9.5 Structural feature claim ledger

The following ledger fixes what is structurally claimed, what is numerically left open, and which support modules carry the load.

Feature	Claim in this paper	Support route	Status
Realized physical interior	derived/module supported	R1–R12	module
Fields and particles	persistence carriers and localized persistence classes	R6–R7	module
Gauge transition, connection, and curvature roles	standing-preserving transport, connection, and holonomy obstruction	GRR1–GRR3	module
Representation and charge-witness rigidity	representation admissibility and charge witness constraints	GRR4; SR3–SR4; PSM3–PSM4	module
Anomaly obstruction and anomaly ledger	anomaly as standing-failure obstruction and survivor-ledger closure	GRR5; SR5; PSM5	module
Gauge-vacuum compatibility	vacuum-compatible residual sector normal form	GRR6; SR6; PSM6	module
Product-sector organization	product gauge-sector compatibility and simultaneous constraint collapse	PSM2; SR1–SR2	module
Standalone SM realization profile	SR7 survivor profile over the admissible gauge arena	SR7 with baseline SM classification/Structura	structural closure
Mass-response layer	mass as response/readout; Higgs-like bridge/fixation carrier; Yukawa-realizability and coupling-slot ledger; mass slots	R9; MGN3–MGN6; L1 Yukawa Equivalence; MGN4; Mass Fixation	module
Numerical readiness status	readiness, forcing, and exactness discipline; not full table closure	MGN8; CNF9; EX1–EX9	status discipline
Charged-lepton parameter module	conditional one-anchor sectoral readout and protected residual localization	MR6–MR7; CLM–CLR; CLS	conditional / open
Full empirical parameter table	not claimed closed here	future sector modules	open

9.6 Standard Model Structural Spine Theorem

Theorem 9.7 (Standalone Standard Model Structural Spine). *Given the AASC kernel, interface layer, realized physical interior, gauge-realization layer, PSM layer, representation-response layer, charge-lattice rigidity, anomaly-ledger rigidity, vacuum-compatibility layer, and the later Standard Model rigidity modules, the admissible physical realization closes to the recovered Standard Model structural tuple*

$$K_{\text{AASC}} + I_{\text{AMetric}} + \mathcal{I}_{\text{phys}} + G_{\text{GRR}} + B_{\text{PSM}} + R_{\text{SR}} + M_{\text{MGN}} + F_{\text{CNF}} \Rightarrow \Sigma_{\text{SM}} / \simeq_{\text{Adm}},$$

where

$$\Sigma_{\text{SM}} = (SU(3)_c \times SU(2)_L \times U(1)_Y, \mathcal{R}_{\text{SM}}, H_{\text{br/fix}}, \mathcal{O}_Y, \mathcal{Q}_{\text{SM}}, \mathcal{A}_{\text{anom}} = 0, \mathcal{V}_{\text{SM}}, \mathcal{M}_{\text{slot}}, \Theta_{\text{mix}}, N_g = 3).$$

Here equality is structural equality up to admissible redescription. It is not equality of all numerical parameter values.

Proof spine. The AASC kernel restricts construction to admissibility-by-standing. The interface layer blocks primitive metric, coordinate, selector, and field assumptions. The realized-physics bridge admits a physical interior only through persistence, localization, continuation, finite propagation, field carriers, particle roles, vacuum support, mass response, and causal ordering. The gauge-realization and PSM layers supply standing-preserving transition organization, connection-like transport comparison, curvature/holonomy residue, representation response, charge compatibility, anomaly obstruction, and gauge-vacuum structure. SR1–SR7 close the admissible gauge-realization survivor profile to the recovered Standard Model gauge product, representation, charge, anomaly, and residual/vacuum skeleton. MGN3–MGN8 and CNF1–CNF9 attach the bridge/fixation, mass-response, Yukawa-compatibility, mixing/phase, and numerical-status layers without promoting unresolved parameter values into structural closure. Exactness then controls any later promotion of values. Therefore the theorem derives the standalone Standard Model structure without asserting full parameter-table closure [1, 6, 43, 85, 86, 87, 88, 89, 90, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 58, 62, 60, 61, 63, 64, 66, 55, 71, 59]. $\square$

9.7 Why this is not reverse engineering

A reverse-engineering derivation would begin with the Standard Model Lagrangian and then reinterpret its features as AASC outputs. The proof spine does not do this. It begins with admissibility, standing, scope, and construction constraints. The standalone Standard Model structure appears only after physical interior, gauge/role carriers, PSM specialization, SR rigidity, and mass-response rules have been licensed.

9.8 Structural certificate

Module output	standalone Standard Model structural occupant
Dependency	kernel; interface; physical interior; gauge realization; PSM; SR rigidity; charge/anomaly/vacuum compatibility; mass-response slots
Used by	numerical-readiness and downstream parameter-sector modules, beginning with charged leptons
Status	structural closure; not a complete derivation of every numerical parameter value

10 Charged-Lepton Parameter-Sector Module

10.1 Module role

The charged-lepton arc is not the completion of the Standard Model parameter table. It is the downstream parameter-sector module after structural closure. Its role is to demonstrate that, once the standalone Standard Model structure and mass-response slots are fixed, a specific numerical sector can be attacked by the numerical-readiness, constructive-forcing, mass-ratio, exactness, and charged-lepton machinery. The module therefore tests the parameter program; it is not the definition of structural closure.

10.2 Role-cell residue

The charged-lepton arc isolates a physically typed mass-shape residue

$$R_\ell^*$$

and a role-projective readout

$$\rho_\ell^{\text{role}}(R_\ell^*) = [\pi_e^{\text{role}} : \pi_\mu^{\text{role}} : \pi_\tau^{\text{role}}].$$

The residue chain identifies the charged-lepton role-cell residue

$$\delta_\ell = \frac{2}{9}.$$

The important point is that this is not an observed muon/tau input. It is a role-cell residue certified upstream in the CLM/CLQ/CLR chain [80, 81, 82, 83].

10.3 Square-root amplitude geometry

Let

$$q_i = \sqrt{\alpha_i}$$

be the square-root response amplitude. The charged-lepton Koide-type invariant is

$$K_\ell = \frac{q_e^2 + q_\mu^2 + q_\tau^2}{(q_e + q_\mu + q_\tau)^2}.$$

The balanced square-root amplitude endpoint gives

$$K_\ell = \frac{2}{3}.$$

In the minimal Hermitian Z_3 -circulant carrier, the carrier eigenvalues have the form

$$q_n = \mu(1 + 2\eta \cos(\delta + 2\pi n/3)).$$

For such a carrier,

$$\sum_n q_n = 3\mu, \quad \sum_n q_n^2 = 3\mu^2(1 + 2\eta^2),$$

so

$$K_\ell = \frac{1 + 2\eta^2}{3}.$$

Thus $K_\ell = 2/3$ forces

$$\eta^2 = \frac{1}{2}.$$

10.4 Boundary-transition carrier

The carrier constructed in CLR v4.6 is

$$\Gamma_\ell(\mu, \eta, \delta) = \mu \begin{pmatrix} 1 & \eta e^{i\delta} & \eta e^{-i\delta} \\ \eta e^{-i\delta} & 1 & \eta e^{i\delta} \\ \eta e^{i\delta} & \eta e^{-i\delta} & 1 \end{pmatrix},$$

with

$$\eta = \frac{1}{\sqrt{2}}, \quad \delta = \frac{2}{9}.$$

The carrier is a square-root amplitude boundary-transition carrier. It is not a Standard Model mass matrix, Hamiltonian, or Yukawa matrix.

The resulting carrier shape is

$$[m_e : m_\mu : m_\tau]_{\text{carrier}} = [1 : 206.7703159727 : 3477.4728371047].$$

10.5 One-anchor metric display

Using the electron pole mass as one metric skin import, the one-anchor display is

$$m_i^{(e)} = m_e^{\text{obs}} \frac{q_i^2}{q_e^2}.$$

This gives

$$m_\mu^{\text{carrier}+e} = 105.659414496 \text{ MeV}, \quad m_\tau^{\text{carrier}+e} = 1776.984970813 \text{ MeV}.$$

Observed muon and tau masses do not enter the construction. They enter only after the readout as diagnostic comparisons.

10.6 Protected pole-boundary correction

CLS then studies the residual between the carrier display and pole comparison as a tensor-to-skin problem. The protected rank-2 correction is

$$D_i^{\text{prot}} = \exp \left[\left(\frac{\alpha}{\pi} \right)^2 \left(-\frac{31}{9} \hat{b}_i^{\text{rad}} + \frac{8}{27} b_i^\theta \right) \right].$$

The corrected display is

$$m_\mu^{\text{prot}+e} \approx 105.658428708 \text{ MeV},$$

$$m_\tau^{\text{prot}+e} \approx 1776.861217971 \text{ MeV}.$$

The residuals become approximately

$$\Delta_\mu \approx +0.000053208 \text{ MeV}, \quad \Delta_\tau \approx +0.001217971 \text{ MeV}.$$

This is a conditional residual-localization improvement: it moves the discrepancy into a sharply constrained protected pole-boundary class. It is not exact pole closure.

10.7 Protected trace remainder

CLS v2.11 separates ordinary scheme displacement, carrier-balance protection, and the remaining protected trace remainder:

$$R_\ell^{\text{prot}} = T_\ell^{\text{trace}} - T_\ell^{\text{scheme}} - T_\ell^{\text{carrier-bal}}.$$

The raw two-loop OS/MS source displacement was tested and excluded as the protected finite trace remainder. The exact open object is therefore

$$\boxed{R_\ell^{\text{prot}}}.$$

10.8 Charged-lepton module certificate

Module output	conditional one-anchor charged-lepton readout; conditional protected pole-boundary residual-localization module; exact trace-remainder obstruction
Dependency	CLM–CLR, numerical-readiness arcs, CLS v2.11, ATS/ZPE trace discipline
Used by	quantitative sectoral test for the SM structural derivation; future trace-remainder work
Status	one-anchor readout achieved conditionally; protected residual-localization module supplied conditionally; exact pole closure open

11 Open Problems and Non-Closure Ledger

11.1 Structural closure status

The standalone Standard Model structure is closed at the structural-tuple level by the proof spine: the recovered object is $\Sigma_{\text{SM}}/\simeq_{\text{Adm}}$, containing the conventional recovered gauge product, representation-response skeleton, Higgs-like bridge/fixation carrier, Yukawa-compatibility ledger, charge lattice, anomaly-neutral survivor ledger, vacuum compatibility, mass-response slots, mixing/phase slots, family-cardinality envelope record, and same-scope structural rival exhaustion.

11.2 Parameter closure status

The empirical parameter table is not closed by the structural theorem. The parameter problem remains sectoral. A sector moves from boundary/open status to derived status only through an admissible numerical module. The charged-lepton module is the first such sectoral module and remains conditional at the pole-closure and zero-anchor-scale layers.

11.3 Exact charged-lepton pole closure

The exact pole-closure object is

$$R_{\ell}^{\text{prot}}.$$

The support CLS arc has narrowed the problem: the missing object is not a carrier, not a coordinate algebra, not a raw OS/MS two-loop scheme displacement, and not an arbitrary finite counterterm. It is a protected, relational, leaf-relative trace remainder.

11.4 Zero-anchor absolute charged-lepton scale

The charged-lepton numerical result uses the electron mass as a skin-level metric import. A zero-anchor derivation would require a charged-lepton leaf-relative scale carrier

$$\Sigma_{\ell, \mathcal{L}} : (A_{\ell}^{\text{scale}}, T_{\ell}^{\text{scale}}, \text{Skin}_{\ell}^{\text{metric}}) \rightarrow S_{\ell, \mathcal{L}}.$$

The metric-fixation and no-global-fixation results prohibit replacing this with a globally privileged mass scale [11, 12]. The zero-anchor scale remains open.

11.5 Downstream parameter-sector roadmap

The parameter program begins after structural closure. The charged-lepton module is the first opened parameter sector. Remaining sectors include:

1. quark mass ratios and pole/running conversion status;
2. CKM mixing angles and CP phase values;
3. PMNS mixing values and neutrino mass splittings;
4. gauge-coupling invariant relations and possible dimensionless overlap constraints;
5. Higgs-sector numerical values;

6. absolute scale anchors and zero-anchor scale closure;
7. dimensionless constants such as α , where admissible invariant standing can be certified.

Each sector requires its own admissible witness, forcing ledger, exactness certificate, and calibration-status audit.

Item	State	Status
AASC kernel	admissibility-by-standing constraint engine	proved / module
Interface	AMetric boundary, scope discipline, no illicit transport	proved / module
Physical interior	boundary-licensed realized physical interior	module supported
Gauge/PSM/SR structure	standalone SM structural occupant	structural closure
Numerical status discipline	readiness, forcing, exactness, and calibration-status machinery	module supported
Charged-lepton one-anchor readout	m_μ, m_τ read from m_e anchor and carrier shape	achieved conditionally
Protected pole-boundary localization	residuals localized into protected trace-remainder class	supplied conditionally; exact remainder open
Exact pole closure	protected trace remainder R_ℓ^{prot} not derived	open
Zero-anchor scale	scale carrier $\Sigma_{\ell, \mathcal{L}}$ not derived	open
Full parameter table	quarks, CKM, PMNS, gauge couplings, Higgs values, α , absolute scales not all closed here	downstream program

12 Consolidated AASC-to-Standard-Model Theorem

Theorem 12.1 (Consolidated Structural Derivation). *The AASC kernel, interface, realized-physics bridge, gauge-realization layer, PSM layer, charge-rigidity layer, anomaly/vacuum compatibility layer, and Standard Model rigidity layer jointly derive the recovered Standard Model structural tuple $\Sigma_{\text{SM}}/\simeq_{\text{Adm}}$ up to admissible redescription and declared boundary/envelope status for unresolved numerical parameters.*

Theorem 12.2 (Consolidated Parameter-Program Status). *The complete Standard Model parameter table is not derived in this manuscript. The charged-lepton arc supplies the downstream parameter-sector module, yielding a conditional one-anchor numerical readout and protected residual localization. Other parameter sectors remain targets for later numerical-forcing and exactness arcs.*

Modular proof. The proof is by module composition.

Kernel. Non-degenerate construction and the kernel of admissibility, structure of admissibility, bivalence, and standing identity closure supply the admissibility-by-standing constraint engine.

Interface. The AMetric boundary, closure by exhaustion, invariant endpoint discipline, and boundary trace fixation block primitive metricity and restrict redescription.

ATS. The anchor–tensor–skin decomposition prevents skin-level display variables from functioning as anchors.

Physical interior. Realized Physics R1–R12 admit a non-vacuous physical interior through persistence, locality, finite transport, fields, particles, gauge boundary trace, mass/inertia, vacuum, time/causality, and final physical interior closure.

ZPE and renormalization. The ZPE arc blocks absolute vacuum offsets and permits only relational, boundary-conditioned response terms.

Gauge realization and PSM. The Gauge Realization arc supplies transition groupoid, connection, curvature/holonomy, representation/charge, anomaly, and vacuum-compatibility structure. The PSM arc licenses the Standard-Model-facing gauge arena without treating the final SM Lagrangian as a premise.

Standard Model rigidity. SR1–SR7 collapse the product, representation, charge, anomaly, and residual/vacuum survivor ledgers into the standalone Standard Model structural occupant, with the earlier Classification and Structura papers retained as baseline occupant/interior closure support.

Mass and numerical discipline. R9, MGN, CNF, MR, and Exactness modules supply mass as deformation/inertial response, numerical readiness, forcing-status discipline, one-anchor reconstruction, and exactness promotion rules.

Charged leptons. CLM–CLR construct the 2/9 charged-lepton role-cell residue, the Z_3 -circulant square-root amplitude carrier, and the conditional one-anchor numerical readout. CLS supplies a conditional protected pole-boundary residual-localization module and localizes the exact remaining obstruction to R_ℓ^{prot} .

The outputs compose without using downstream observed muon/tau masses as construction inputs, without primitive metricity, and without global scale fixation. The open parameter objects are explicitly recorded and do not invalidate the completed structural derivation. This establishes the consolidated claim. $\square$

13 Anticipated Objections and Responses

Objection	Spine response
“You have derived a role schema, not the Standard Model.”	The recovered tuple is displayed explicitly: $SU(3)_c \times SU(2)_L \times U(1)_Y$, chiral representation-response skeleton, Higgs-like bridge/fixation carrier, Yukawa-compatibility ledger, charge lattice, anomaly-neutral survivor ledger, vacuum-compatible residual form, mass/mixing slots, and $N_g = 3$ family-cardinality envelope record. Role language is the derivation medium, not a substitute for the recovered physics object; the field-content table gives the conventional representation slots explicitly.
“You have derived gauge-role structure, not $SU(3)_c \times SU(2)_L \times U(1)_Y$.”	The gauge proof does not stop at the generic role. The Gauge Realization ladder forces transition organization, transport comparison, curvature/holonomy residue, representation response, charge-witness rigidity, anomaly neutrality, and gauge-vacuum compatibility. Composed with PSM, SR, and charge-lattice modules, this recovers the conventional Standard Model gauge-product presentation.
“Extra gauge factors are counterexamples.”	An extra gauge factor is a same-scope counterexample only if it preserves the same representation, charge, anomaly, witness, vacuum, transport, and boundary ledgers. If it preserves them, it factors through the same quotient; if it changes any ledger, it is larger-domain load rather than a same-scope rival.
“You have not derived g_1, g_2, g_3 .”	Correct. Gauge-coupling numerical values are not part of the structural theorem. The gauge support theorems classify raw gauge-coupling values as moduli or skin-level chart data unless additional dynamical, boundary, fixed-point, or empirical structure is supplied. The structural theorem derives the gauge product and admissibility ledger, not the numerical coupling table.
“Gauge symmetry is being made ontic.”	No. Gauge symmetry is treated as representational bookkeeping and admissible redescription structure. The ontic-gauge reading is rejected; gauge-related presentations are not distinct physical occupants.
“A Higgs-like bridge is not the full SM Higgs sector.”	Correct. The structural tuple uses $H_{\text{br/fix}}$ as the admissible bridge/fixation carrier required for order-to-deformation mediation and mass-response compatibility over the recovered skeleton. MGN3 blocks vev, potential, normalization, Yukawa, mass-matrix, and mixing imports at the bridge stage. Higgs potential, vev, scalar mass, self-coupling, weak-scale values, and full scalar-sector numerical data remain downstream unless separately certified.

Objection	Spine response
“Yukawa slots are not Yukawa matrices.”	Correct. $\mathcal{O}_Y$ is the mass-response coupling-slot/Yukawa-compatibility ledger, not the numerical Yukawa eigenvalue table. Charged-sector slots are structurally recorded; neutrino-sector options are explicitly envelope/sector-dependent and not silently imported.
“If $N_g = 3$ is not a pole mass or coupling, why is it in the structural tuple?”	Because changing family cardinality changes the representation-family ledger, mixing-coordinate dimensions, Yukawa-compatibility structure, and boundary bookkeeping. It is a structural envelope record, not a raw numerical parameter. The proof spine includes it in Σ_{SM} and forbids treating it as a freely adjustable same-scope multiplicity.
“You have not derived every numerical parameter, so you have not derived the Standard Model.”	This objection conflates structure with numerical instantiation. The theorem derives the standalone Standard Model structural occupant: gauge-product and product-sector organization, representation-response roles, charge lattice, anomaly-compatible survivor profile, vacuum compatibility, mass-response slots, mixing/phase slots, and same-scope structural rival exhaustion. The empirical numerical table is a downstream parameter-derivation problem. The charged-lepton arc is the parameter-sector module in that program.
“The charged-lepton module is incomplete, so the structural theorem fails.”	The charged-lepton module is not a premise of structural closure. It is downstream of structural closure. Its remaining pole-boundary and zero-anchor-scale obstructions limit the parameter claim, not the structural derivation.
“Earlier AASC papers classified residuals as boundary data.”	Correct within their earlier fixed residual scope. The present spine retains those results as no-selector baselines. Later arcs supersede them only where they supply new admissible witnesses, carriers, transport structures, quotient collapses, exactness certificates, or reconstruction rules.
“The paper assumes the Standard Model.”	The chain begins with admissibility, standing, AMetric/interface discipline, physical realization, gauge-realization, and PSM bridge layers. The standalone Standard Model structure appears only after the SR rigidity sequence and prior classification baselines; it is not inserted as a starting Lagrangian.
“Kernel necessity alone cannot force the Standard Model.”	Correct. Kernel necessity fixes the admissibility conditions for non-degenerate construction. The Standard Model tuple is forced only after that kernel is instantiated inside the declared realized physical envelope and composed with the physical-realization, gauge-realization, PSM, SR, and mass-response modules. The claim is therefore kernel-plus-physical-envelope necessity, not an unconstrained inference from the kernel alone.
“Metricity is smuggled in.”	AMetric, metric-fixation, and no-global-fixation results block primitive metricity and global scale fixation. Metric skin is explicitly separated from tensor standing.

Objection	Spine response
“The charged-lepton readout uses muon/tau data.”	Observed m_μ, m_τ are diagnostic comparisons only. The construction uses 2/9, the carrier, and the electron skin anchor.
“The protected pole correction is fitted.”	The protected tensor is stated with conditional status. The exact protected trace remainder is open. Residual-derived coefficients are not certified as final exact closure.
“The Standard Model derivation is incomplete.”	Correct only if “derivation” is interpreted as full numerical closure of every parameter. The claim here is structural derivation of the standalone SM occupant plus a first charged-lepton parameter-sector module.
“The dependency set is too large.”	This paper is the dependency-compressed spine. Module certificates specify which support results are load-bearing and which claims are structural, parameter-level, conditional, or open.
“AASC replaces physical dynamics with trajectory language.”	No. AASC trajectory theory concerns finite fixed-scope construction histories and their standing-preservation geometry. Physical dynamics—Hamiltonian propagation, gauge transport, RG flow, amplitudes, vacuum-sector QFT, and mass response—enter through physical modules. The trajectory layer governs whether those modules are used coherently under admissibility and standing constraints.
“The result is merely philosophical.”	The proof spine ties AASC to physics through explicit instantiation bridges: gauge-transition groupoid structure, ZPE quotient discipline, Higgs/order-to-deformation mediation, Yukawa coupling-witness topology, and neutrino boundary-kernel reconstruction. These are physical-domain applications of standing-preservation discipline, not free philosophical analogies.
“The ZPE discussion ignores standard QFT renormalization.”	The ZPE bridge is a quotient-identification result: the AASC standing quotient in the homogeneous vacuum-energy sector aligns with the renormalized-QFT quotient that removes absolute scalar offsets from standalone predictive status. The result does not solve the cosmological-constant matching problem and does not introduce new vacuum dynamics.
“ZPE or trace terms are arbitrary.”	ZPE support blocks absolute offsets and permits only relational boundary-conditioned response. CLS records the exact protected trace remainder as open rather than inserting a finite counterterm.

14 Conclusion

The strongest closed structural statement is:

AASC gives a first-principles derivation of the recovered standalone Standard Model structural tuple $\Sigma_{\text{SM}}/\simeq_{\text{Adm}}$.

The strongest quantitative statement presently available is:

The charged-lepton arc gives the downstream parameter-sector module: a conditional one-anchor numerical readout with protected pole-boundary residual localization. Exact pole closure and zero-anchor absolute scale remain open.

Structural completion is therefore not parameter completion. The former is the result of this proof spine. The latter is the downstream numerical program exemplified by the charged-lepton arc and continued sector by sector.

The constructional vocabulary used throughout the proof spine is not a substitute for physical dynamics. The spine uses trajectory language to track standing-preserving construction histories; physical dynamics enter through the physical modules themselves. Gauge transport, Hamiltonian propagation, RG flow, vacuum renormalization, Higgs-like bridge mediation, Yukawa compatibility, and mass-response structure remain physical-domain objects. AASC supplies the admissibility discipline under which those objects may be composed without changing the standing quotient, importing selectors, or repairing failed lineages.

The resulting final statement is:

AASC derives the recovered standalone Standard Model structural tuple, including explicit representation slots and Yukawa-realizability/coupling-slot structure. The full empirical parameter table remains a downstream sectoral program.

15 Operational Glossary

Admissibility	The legality constraint on construction and redescription.
Standing	The global binary property preserved by admissible construction and reference.
Reference	Standing-preserving admissible redescription.
Scope	The bounded domain in which admissibility and standing are evaluated.
AMetric boundary	The theorem layer blocking primitive metricity before admissible fixation.
Anchor	Standing source licensing a domain or scale of comparison.
Tensor	Load-bearing carrier that transports or organizes standing content.
Skin	Display representation of tensor output, such as MeV masses, coordinates, or scheme variables.

Physical interior	Boundary-licensed realized domain of physical persistence and carriers.
Role carrier	Standing-preserving organization of physical roles under admissible transport.
Standalone Standard Model structure	The structural tuple $\Sigma_{\text{SM}}/ \simeq_{\text{Adm}}$ containing the recovered recovered gauge product, representation-response skeleton, Higgs-like bridge/fixation carrier, Yukawa-compatibility ledger, charge lattice, anomaly-neutral survivor ledger, vacuum compatibility, mass-response slots, mixing/phase slots, family-cardinality envelope record, and same-scope structural rival exhaustion.
One-anchor numerical readout	A dimensionless shape reconstruction displayed by importing one metric skin scale; it is not a zero-anchor absolute-scale derivation.
Protected trace remainder	The remaining finite pole-boundary trace object required for exact charged-lepton pole closure.
Parameter-derivation program	The downstream sector-by-sector numerical program that begins after standalone SM structural closure.

16 Module Certificates

16.1 Kernel module

Input	minimal construction setting with admissibility, standing, reference, scope
Output	admissibility-by-standing constraint engine
Support	Non-Degenerate Construction and Kernel of Admissibility; Structure of Admissibility; Bivalence; Standing Identity Closure
Used by	every downstream module
Status	proved / module

16.2 Interface module

Input	kernel constraints
Output	AMetric boundary; scope and redescription discipline; endpoint and boundary trace rules
Support	AMetric Boundary; Closure by Exhaustion; Invariant Endpoint; Boundary Trace
Used by	physical interior, gauge, metric scale, pole trace
Status	proved / module

16.3 Physical realization module

Input	kernel + interface
Output	boundary-licensed realized physical interior
Support	Realized Physics R1–R12
Used by	gauge, fields, particles, mass, vacuum, time, SM realization
Status	module supported

16.4 ZPE/trace module

Input	physical interior + renormalization/standing discipline
Output	absolute vacuum offsets blocked; relational boundary response allowed
Support	ZPE arc
Used by	protected trace-remainder and pole-boundary residual discipline
Status	proved / module

16.5 Gauge/role and SM structural module

Input	physical interior + boundary trace + gauge-realization/PSM bridge
Output	role-stable gauge carrier architecture and standalone SM structural occupant
Support	R8; Gauge Realization; PSM; SR1–SR7; gauge-overlap closure; gauge-coupling moduli; SM Classification; Structural
Used by	structural derivation theorem and downstream parameter-sector roles
Status	structural closure / module

16.6 Mass/numerical module

Input	physical role carriers
Output	mass as deformation/inertial response; numerical readiness; ratio/shape reconstruction
Support	R9; MGN arc; CNF arc; MR arc
Used by	charged-lepton numerical reduction
Status	module supported

16.7 Charged-lepton module

Input	mass-response and numerical readiness modules
Output	$2/9$ residue, Z_3 -circulant carrier, one-anchor numerical readout, conditional protected pole-boundary residual-localization module
Support	CLM–CLR v4.6; CLS v2.11
Used by	first parameter-sector module after structural closure
Status	one-anchor readout achieved conditionally; exact pole closure open

17 Charged-Lepton Numerical Details

The carrier eigenvalues may be written as

$$q_n = \mu \left(1 + \sqrt{2} \cos \left(\frac{2}{9} + \frac{2\pi n}{3} \right) \right),$$

with the positive charged-lepton chamber selected by the symbolic role-order certificate. Squaring and projectivizing yields

$$[1 : 206.7703159727 : 3477.4728371047].$$

The one-anchor display is

$$m_i^{(e)} = m_e^{\text{obs}} \frac{q_i^2}{q_e^2}.$$

The protected tensor uses

$$D_i^{\text{prot}} = \exp \left[\left(\frac{\alpha}{\pi} \right)^2 \left(-\frac{31}{9} \widehat{b}_i^{\text{rad}} + \frac{8}{27} b_i^\theta \right) \right].$$

This gives

$$m_\mu^{\text{prot}+e} \approx 105.658428708 \text{ MeV}, \quad m_\tau^{\text{prot}+e} \approx 1776.861217971 \text{ MeV}.$$

Residuals relative to the comparison values are

$$\Delta_\mu \approx +5.3208 \times 10^{-5} \text{ MeV}, \quad \Delta_\tau \approx +1.217971 \times 10^{-3} \text{ MeV}.$$

The muon residual remains above the precision uncertainty of the muon mass. Therefore exact pole closure is not claimed.

18 Open Problem Ledger

Open object	Meaning	Next theorem
R_ℓ^{prot}	protected finite trace-remainder vector for exact charged-lepton pole closure	trace-remainder theorem
$\Sigma_{\ell, \mathcal{L}}$	leaf-relative charged-lepton scale carrier replacing electron skin anchor	scale-carrier theorem
Full electroweak scale bridge	exact derivation of electroweak/Higgs-like scale input where not already module-certified	bridge theorem
Remaining SM numerical constants	gauge/mixing/mass constants not closed by the present modules	sector-specific CNF/MGN closure
Neutrino/quark numerical modules	not compressed in this charged-lepton endpoint paper except through structural status	separate numerical spine

19 Claim-Level Support Ledger

This appendix records the individual load-bearing claims in the derivation. The ledger is intentionally conservative: it lists not only the support source but also the kind of support supplied and the remaining vulnerability, if any.

Claim	Primary support	Spine use	Status
Non-degenerate construction requires admissibility-by-standing	Non-Degenerate Construction and Kernel of Admissibility; Structure of Admissibility; Bivalence; Standing Identity Closure	root legality condition for all later modules	proved / module
Metricity is not primitive	AMetric Boundary; Metric Fixation; No Global Metric Fixation	prevents smuggling metric scale, coordinates, or dynamical variables into the kernel	proved / module
Redescription must preserve standing and scope	Structure of Admissibility; Closure by Exhaustion; Boundary Trace	blocks post-selection and silent transfer between skins/leaves	proved / module
Physical interior is boundary-licensed, not primitive	Realized Physics R1–R12	bridge from AASC to physical structure	module supported
Vacuum offsets are not predictions	ZPE arc	blocks absolute ZPE and arbitrary finite offset claims	proved / module
Relational boundary responses may carry standing	ZPE arc; Boundary Trace; R12	permits trace/pole response only when relational and witnessed	module supported
Gauge-like carriers are role-stable transport carriers	R8; Gauge Realization; Boundary Trace; Gauge Coupling Parameters	supports gauge/role layer of SM structure	module supported
Recovered standalone Standard Model structural tuple is the support-classification survivor	SR7; Gauge Realization; PSM; A Classification Theorem; Structura	central structural spine claim	structural closure
Mass is response/inertia readout	R9; MGN; MR arc	prevents primitive mass scalar assumption	module supported
One-anchor reconstruction is lawful display, not zero-anchor derivation	MR6; ATS; Metric Fixation	classifies electron anchor as skin import	module supported

Claim	Primary support	Spine use	Status
Charged-lepton role residue is $2/9$	CLM; CLQ; CLR	phase/carrier input for first parameter module	module / conditional
Carrier radius is fixed by $K_\ell = 2/3$	CLQ; CLR	forces $\eta^2 = 1/2$	achieved in module
Z_3 -circulant carrier gives one-anchor readout	CLR v4.6	charged-lepton first parameter-sector endpoint	achieved conditional on same-coordinate carrier admission
Observed m_μ, m_τ are not construction inputs	CLN; CLR; CLS	calibration firewall	achieved by dependency graph
Protected pole-boundary correction improves residuals	CLS v2.4–v2.11	shows nontrivial residual localization and correction class	conditional; exact pole remainder open
Raw two-loop OS/MS is not the protected remainder	CLS v2.10–v2.11	excludes a tempting but wrong closure route	achieved within CLS
Exact pole closure requires R_ℓ^{prot}	CLS v2.11	names final charged-lepton pole obstruction	open
Zero-anchor absolute scale requires $\Sigma_{\ell, \mathcal{L}}$	ATS; Metric Fixation; MR6/MGN3	separates shape closure from absolute scale derivation	open

20 Compression Map from Distributed Corpus to Spine

The master spine compresses the distributed corpus by treating each arc as a theorem module. The compression rule is strict: a support theorem may be compressed only to an output that is explicitly used by a downstream module. The following table records the compression.

Corpus region	Compressed output	Downstream use
AASC kernel papers	non-degenerate construction, admissibility-by-standing, bivalence, standing identity, construction legality	all modules
AMetric/interface papers	no primitive metricity, no illicit scope transport, endpoint/trace discipline	physical bridge, metric/scale restrictions, pole trace
ATS paper	anchor/tensor/skin separation	scale import, charged-lepton carrier vs pole skin, ZPE/trace classification
Realized Physics R1–R12	realized physical interior with locality, fields, particles, gauge boundary, mass/inertia, vacuum, time	physical basis for SM structural carrier modules
ZPE arc	absolute offsets blocked; relational boundary response permitted	finite trace-subtraction admissibility
Gauge/PSM/SR and SM classification	standalone Standard Model structural occupant	central SM spine theorem
Numerical-readiness arcs	readiness, residue admission, one-anchor reconstruction, exactness discipline	charged-lepton reduction and parameter-status discipline
CLM–CLR	charged-lepton residue, square-root amplitude geometry, Z_3 -circulant carrier, one-anchor numerical readout	first parameter-sector module
CLS	conditional protected pole-boundary residual localization, raw OS/MS exclusion, exact remainder obstruction	residual endpoint and future theorem target

21 Reader Navigation Guide

A reader who wants to audit the spine should inspect it in the following order.

1. Verify the kernel: admissibility, standing, reference, construction, and bivalence.
2. Verify that the AMetric/interface layer blocks primitive metric and scale assumptions.

3. Verify the Realized Physics bridge: physical structure is admitted as boundary-licensed persistence and tensor organization.
4. Verify that gauge/role and mass-response modules are not inserted before the physical interior is licensed.
5. Verify the Gauge Realization, PSM, and SR structural modules separately from the charged-lepton numerical module.
6. Verify that the charged-lepton numerical module is a first parameter-sector module and does not use observed muon or tau masses as inputs.
7. Verify that the open objects R_ℓ^{prot} and $\Sigma_{\ell, \mathcal{L}}$ are not silently treated as closed.

This navigation order is also the recommended audit order for external review.

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